Data Communication and Networking

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Outline

1.1 Data Communications

1.2 Networks

Chapter 1: Objective

- The first section introduces data communications and defines their components and the types of data exchanged. It also shows how different types of data are represented and how data is flowed through the network.

- The second section introduces networks and defines their criteria and structures. It introduces four different network topologies.

DATA COMMUNICATIONS: Introduction

•When we communicate, we are sharing information. This sharing can be local or remote. The term telecommunication, which includes telephony, telegraph, and television, means communication at a distance.

•Data communications are the exchange of data between two devices via some form of transmission media.

Introduction

- The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- For data communication to occur, communicating devices must be part of communication system.
- Communication system is made up of combination of hardware(Physical equipment) and software(Programs)

Fundamental Characteristics of Data communication

- *Delivery*

  -- Deliver data to correct Destination.

- *Accuracy*

  -- Must deliver data accurately i.e no alteration of data in transmission.

- *Timeliness*

  --Deliver data in timely manner.

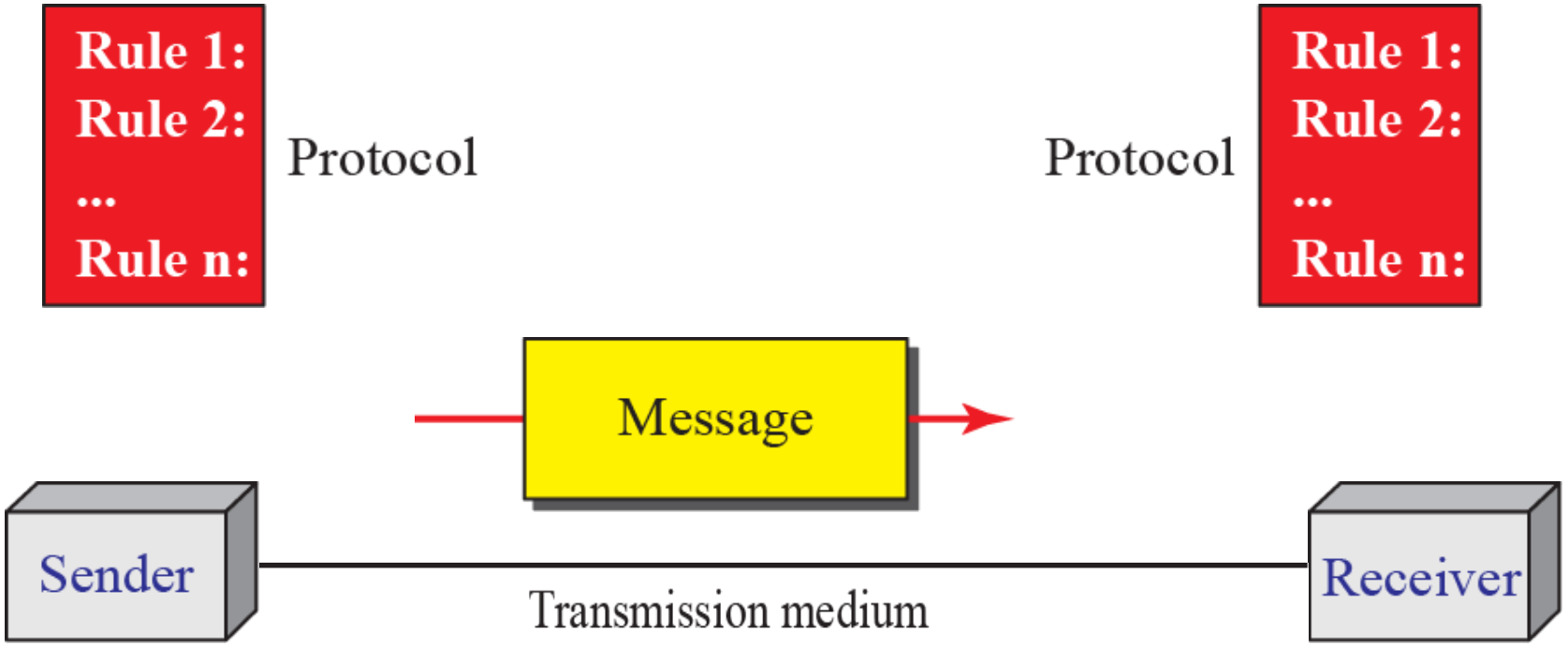
- *Jitter*

-- Refers to variation in packet arrival time.

Components

A data communications system has five components

Five components of data communication



Components are

1.Message:

Data or information to be communicated. Eg. Text, nos, images, audio or video

2. *Sender :* Device which sends data message
3. *Receiver:* Device that receives messages
4. *Transmission medium:*

Physical path by which message is transmitted from sender to receiver. Eg. Twisted pair cable, Coaxial cable. Radio waves, Fiber-optic cable etc.,

5. *Protocol :*

Set of rules that governs data communication. It represents agreement between communicating devices. Without a protocol two devices may be connected but not communicating.

Data representation

- Information is in the form of
    1. Text
    2. Numbers
    3. Images
    4. Audio
    5. Video

Text

- Represented as a bit pattern, a sequence of bits
- Different sets of bit patterns have been designed to represent text symbols.
- Each set is called a **code** and the process of representing symbols is called **coding**.
- *Unicode* –uses 32 bits to represent a character or symbol in any language in the world.
- *ASCII(Basic latin)* - 7 bit code, designed to provide code for 128 symbols. Now constitutes first 127 characters in Unicode.

Numbers

- Numbers are also represented by bit patterns.
- Numbers are directly converted to binary form to simplify mathematical operations.

Images

- Images are also represented by bit patterns.
- Image is composed of a matrix of pixels (picture elements), where each pixel is a small dot
- The size of pixel depends on resolution.
- For ex, an image is divided into 1000 pixels or 10,000 pixels.
- In the second case, better representation of image (better resolution), but more memory is needed to store the image

Image

- Each pixel is assigned a pattern. Size and value depends on the image.
- Black White image - 1 bit pattern
- Ex: chess board

Gray scale- can use **2 bit pattern**

Color image - RGB or YCM, intensity of each color is measured and a bit pattern is assigned

Audio and video

- Audio is recording and broadcasting of sound or music.
- It is continuous, not discrete.
- Video is recording and broadcasting of picture or movie.
- Video can be continuous (ex: TV Camera) entity or combination of images (discrete), each a discrete entity arranged to convey the idea of motion.

Data Flow

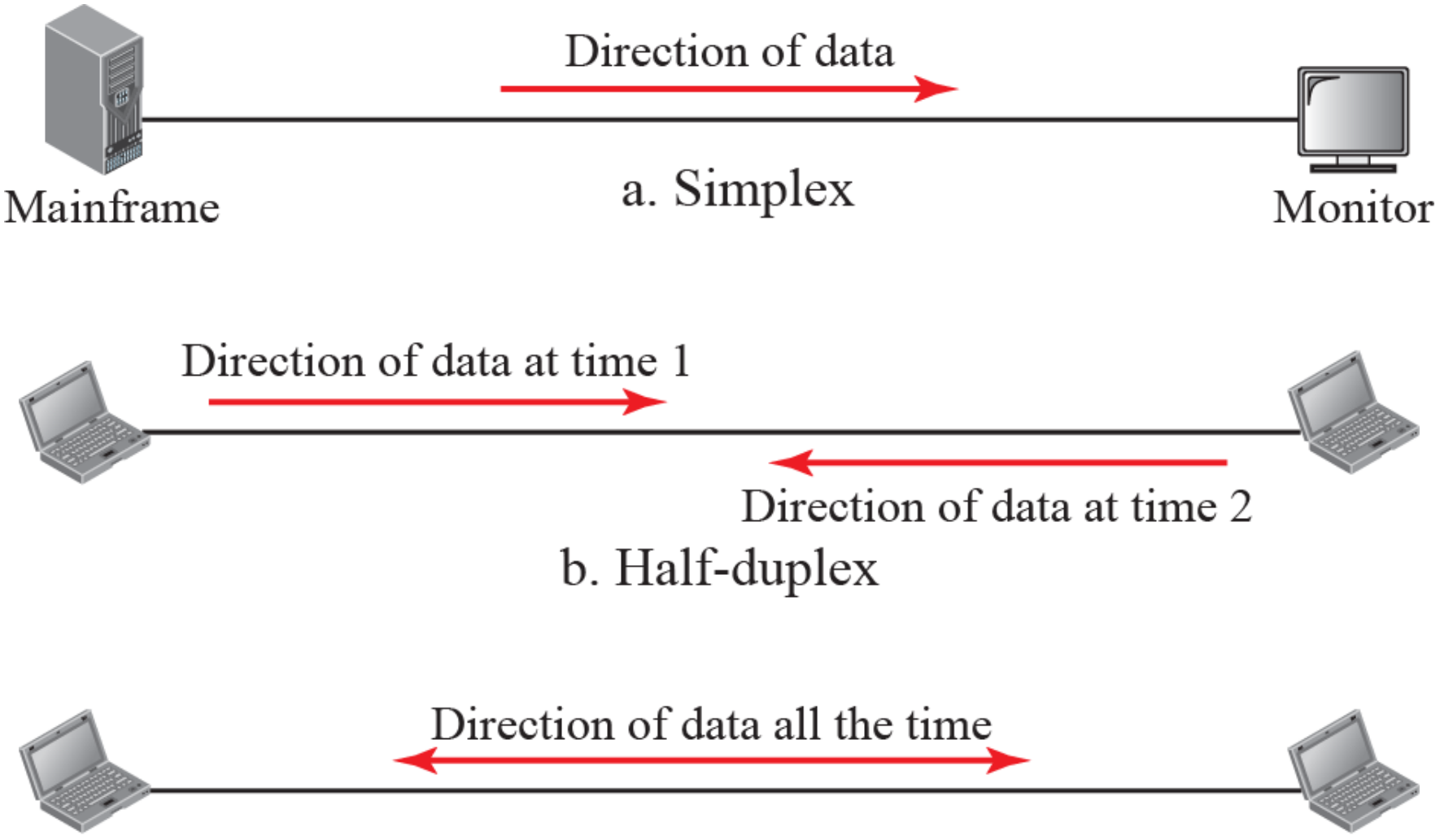
- Communication between two devices can be

i)Simplex

ii)Half-duplex

iii)Full-duplex.

Data flow


Direction of data
Mainframe
Monitor
a. Simplex
Direction of data at time 1
Direction of data at time 2
b. Half-duplex
Direction of data all the time
c. Full-duplex

NETWORKS

- A network is the **interconnection of a set of devices** capable of communication.

- In this definition, a device can be a host such as a large computer, desktop, laptop, workstation, cellular phone, or security system.

- A device in this definition can also be a connecting device such as a router, a switch, a modem that changes the form of data, and so on.

Network Criteria

- A network must be able to meet a certain number of criteria.

- The most important of these are

  Performance

  Reliability

  Security

Performance

- It is measured in many ways

  1. Transit time: Amount of time required for a message to travel from one device to another.

  2. Response time: Time elapsed between enquiry and response.

- Performance depends on **numbers of users, medium, capability of connected hardware and efficiency of software**
- Is evaluated by networking metrics

  1. Throughput
  2. Delay

Reliability and Security

- Network reliability is measured by
  1. Frequency of failure.
  2. Time it takes to recover from failure.

Security issue includes

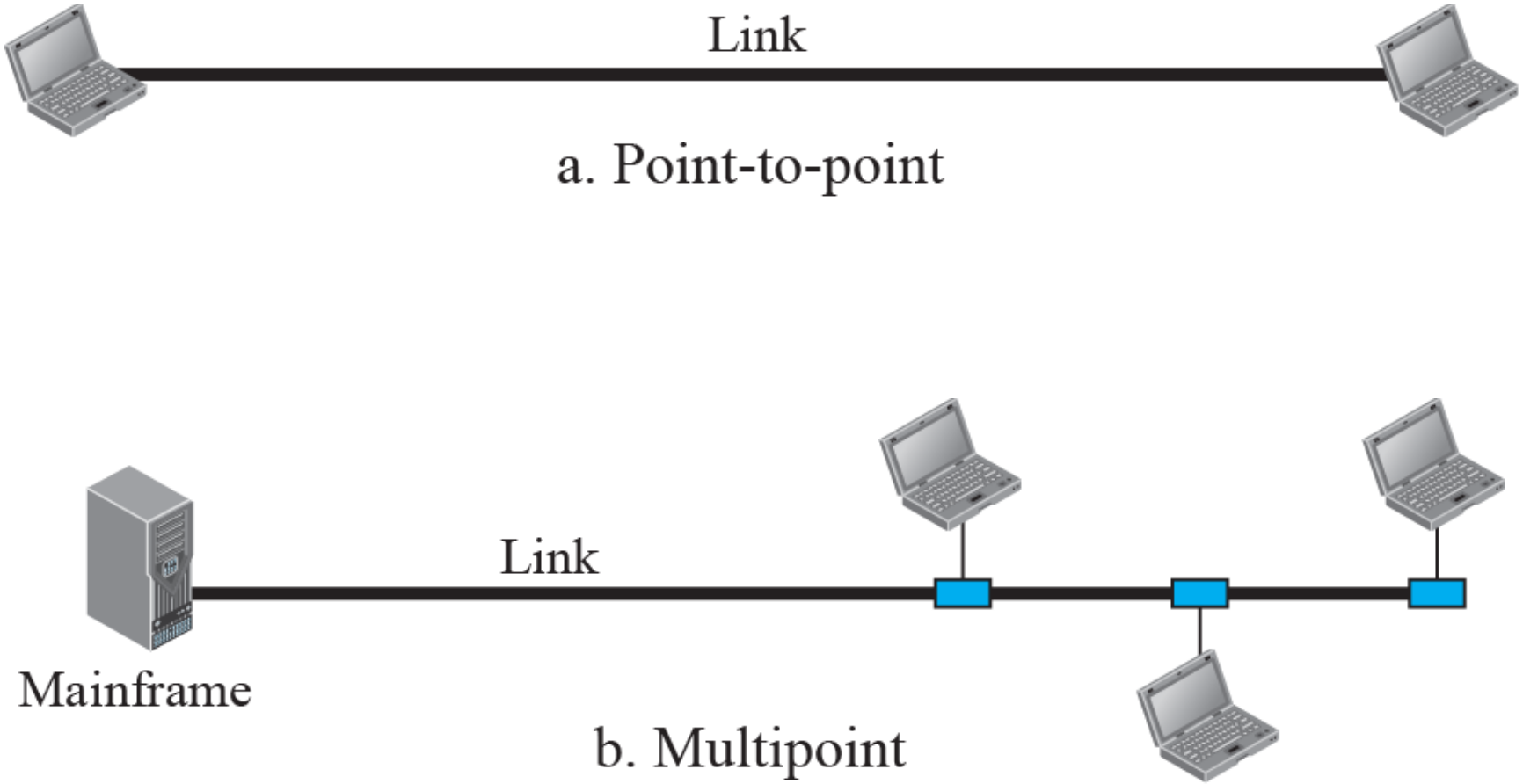
1. Protecting data from damage.
2. Protecting data from unauthorized access.
3. Implementing policies and procedures for recovery from breaches and data losses.

Physical structure

- A network is two or more devices connected through links.
- A link is a communication pathway that transfers data from one device to another.
- For communication to occur,

the two devices must be connected in some way to same link at the same time.

Types of connection



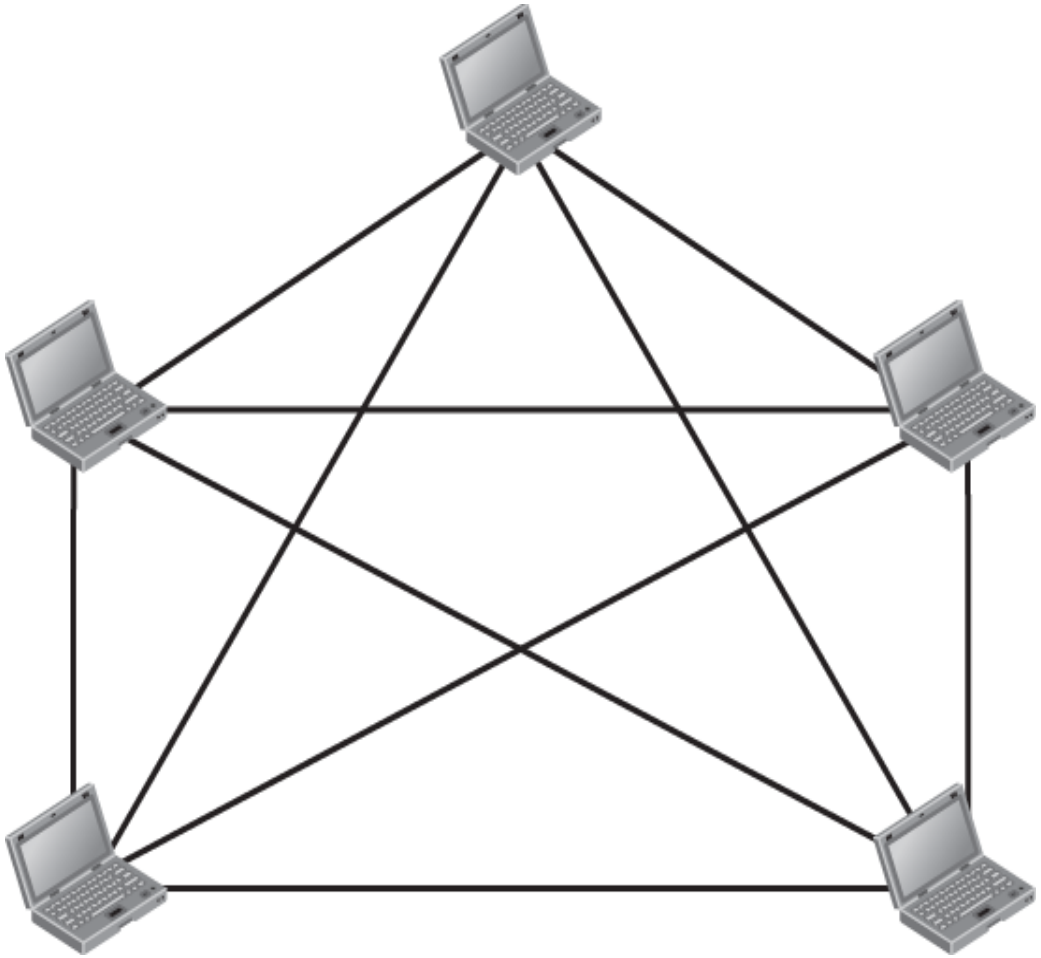
Physical topology

- Refers to the way in which a network is laid out physically.

- Topology of a network is the geometric representation of the relationship of all the links and linking devices(nodes) to one another.

A fully-connected mesh topology

n = 5



10 links

Mesh topology

- Every device has a dedicated point-to-point link to every other device.
- Dedicated means link carries traffic between only two devices it connects.
- To find the number of physical links in a fully connected(Each node must be connected to every other node) mesh network with n nodes,

  Node 1 must be connected to n-1 nodes

  Node 2 must be connected to n-1 nodes

  Node n must be connected to n-1 nodes

Mesh Topology

Therefore,

we need n(n-1) physical links.

- If each physical link allows communication in both the directions, then we can divide the number of links by 2

  Therefore we need n(n-1)/2 duplex- mode links.

- To accommodate many links, every device on the network must have (n-1) I/O ports.

Advantages

1. Use of dedicated link guarantees that each connection carries its data load.
2. Robust . If one link becomes unusable, it does not incapacitate the entire system.
3. Privacy or Security: because message is carried along dedicated link. Physical boundaries prevent others from gaining access to message.
4. Fault identification and isolation is easy. Traffic can be routed to avoid links with suspected problem.

- Easy for the network operator to discover the location of fault & its causes and to find solutions.

Disadvantage

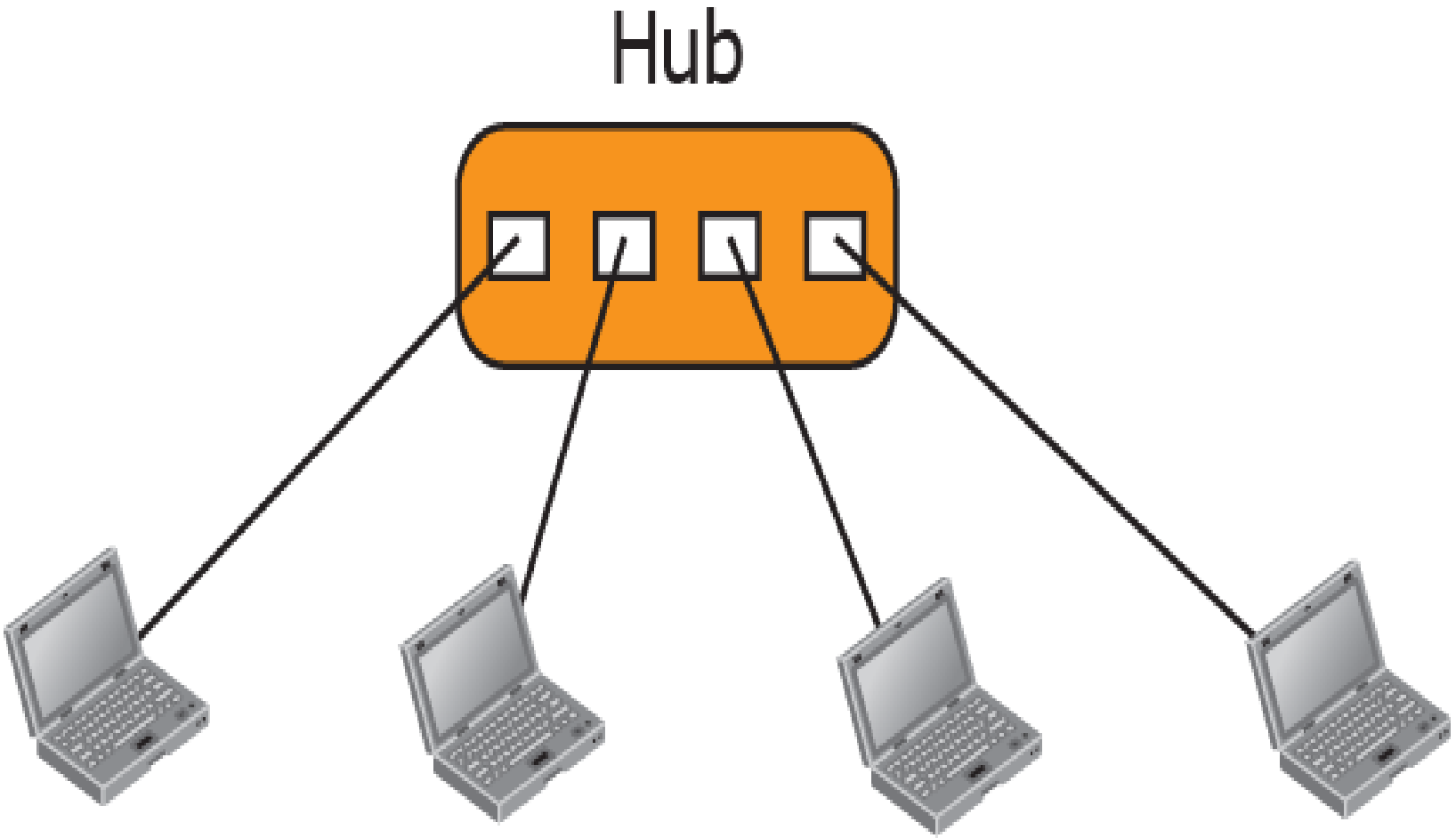
Related to amount of cabling and number of I/O ports required,

1. Installation and reconnection are difficult.
2. Bulk of wiring can be greater than available space.
3. Hardware required to connect each link (I/O ports and cables) are expensive.

For these reasons, Mesh topology is implemented in a limited fashion.

- *Practical example of Mesh topology is the connection of Telephone regional offices in which each regional office need to be connected to every other regional office*.

A star topology



Star topology

- Each device has dedicated point-to-point link only to central controller called *HUB*
- Devices are not directly connected to one another.
- Controller acts as an *Exchange* .
- If one device wants to send data to another, it sends the data to controller which then relays data to other connected device.

Advantages

1. Less expensive.
2. Each device needs only one link and one I/O port to connect to any number of devices.
3. Easy to install and reconfigure.
4. Less cabling. Addition, deletion, moving of devices involves only one link.
5. Robustness i.e if one link fails, only that link is affected.
6. Easy fault identification and fault isolation .Hub can monitor link problems and bypass defective links.

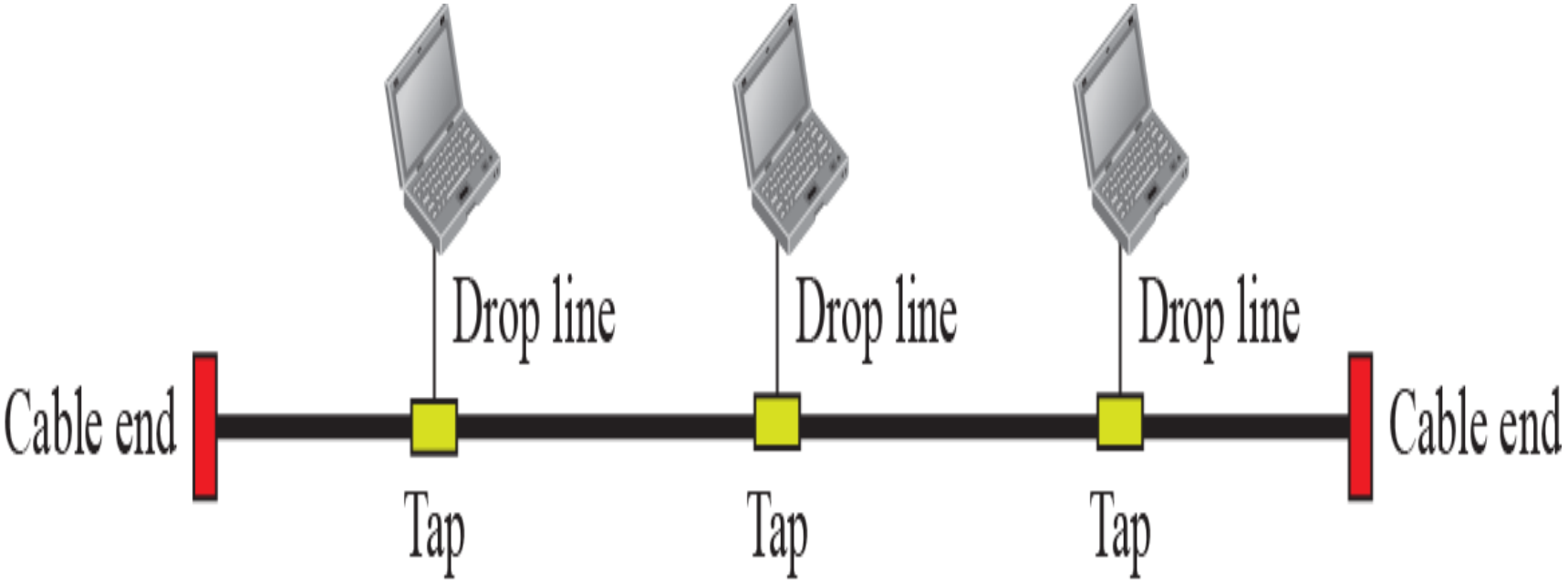
Disadvantages

1. If Hub goes down, entire system is dead.
2. As each node must be linked to hub, more cabling is required than bus and ring topology.

Used

1. In LANS.
2. High Speed LANs use star topology with central hub.

A bus topology



Bus Topology

- Is multipoint.
- One long cable acts as a backbone to link all the devices in a network.
- Nodes are connected to bus cable by drop lines and taps.
- Drop line is connection running between device and main cable.
- Tap is a connector to create a contact with metallic core.

Continued

- As signal travels along the backbone, some of the energy is transformed into heat and as it travels farther and farther, signal becomes weaker.

☐ Therefore, there is a limit on the number of taps a bus can support and the distance between those taps.

Advantages

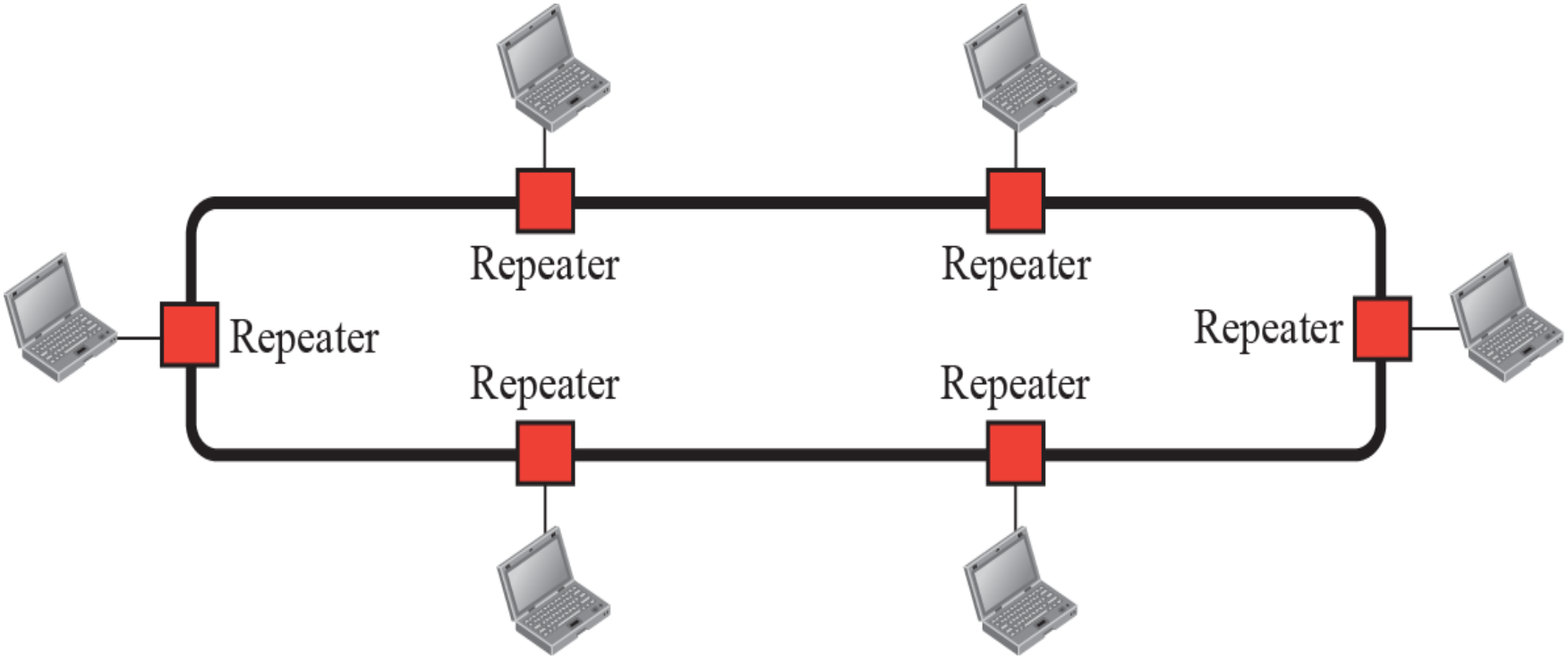
1. Installation is easy.
2. Bus uses less cabling as only the backbone cable stretches.

Disadvantages

1. Fault isolation and reconnection is difficult.
2. Difficult to add new devices.
3. Signal reflection at taps can cause degradation in quality, which is controlled by limiting number and spacing of devices connected to the length of the cable.
4. Break in cable stops all transmission.

- Traditional LAN's used bus topology.

A ring topology



Ring topology

- Each device has dedicated point-to-point connection with only two devices on either side of it.
- Signal passed along the ring in only one direction from device to device until it reaches destination.
- Each device in ring incorporates a repeater.
- When the device receives a signal intended for another device, it regenerates the bits and passes them along

Advantages

1. Easy to install and reconfigure.
2. Fault isolation is simplified.

   ☐ In a ring, a signal is circulating at all the times. If one device does not receive signal within a specified time, it issues alarm and alerts network operator to the problem & its location.

Disadvantage

1. Unidirectional traffic.
2. A break in a ring can disable the entire network.

- Can be solved by using a dual ring.

- Was very popular when IBM introduced its LAN token ring.
- High speed LAN's made this topology less popular.

Chapter 2: Outline

2.1 Protocol Layering

2.2 TCP/IP Protocol Suite

2.3 OSI Model

Chapter 2: Objective

- The first section introduces the **concept of protocol layering** using two scenarios.
- The **two principles** upon which the protocol layering is based
- The first principle dictates that each layer **needs to have two opposite tasks.**
- The second principle dictates that the **corresponding layers should be identical.**

Chapter 2: Objective (*continued*)

- The second section discusses the **five layers** of the TCP/IP protocol suite.

- The third section gives a brief discussion of the OSI model. This model was **never implemented in practice**, but a brief discussion of the model and its comparison with the TCP/IP protocol suite may be useful to better understand the TCP/IP protocol suite.

PROTOCOL LAYERING

- A word we hear all the time when we talk about the Internet is protocol.

- A protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.

- When communication is simple, we may need **only one simple protocol;** when the communication is complex, **we need a protocol at each layer**, or protocol layering.

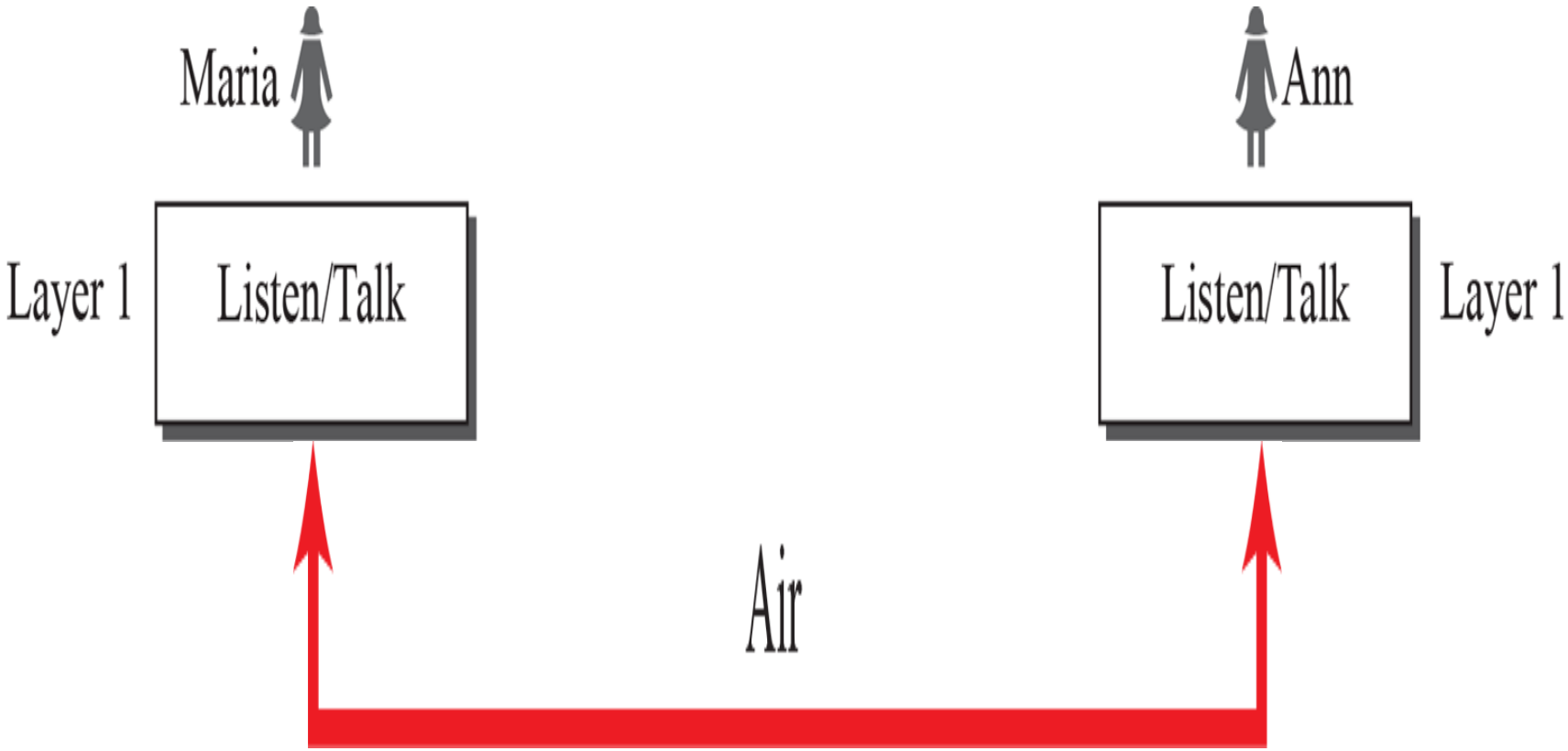
Scenarios

Let us develop two simple scenarios to better understand the need for protocol layering.

In the first scenario, communication is so simple that it can occur in only one layer.

In the second, the communication between Maria and Ann takes place in three layers.

A single-layer protocol



Scenario-1

- Assume that Maria & Ann are neighbors with lot of common ideas.
- Communication between them is **face to face and in same language.**
- Set of rules to be followed

i) They should **greet each other** when they meet.

i) They should **confine their vocabulary** to the level of their friendship.

iii) When other party is speaking, first party should **refrain** from speaking.

iv) Conversation should be a **dialog** [Both party should talk about an issue] not a monolog.

v) They should **exchange nice words** when they leave.

Scenario-2

- Assume that Ann is **offered** a higher-level position in her company but moved to another branch located in a city far from Maria.
- Assume that they have come up with an innovative project to start a new business when they retire.
- Decided to continue their conversation through postal mail.

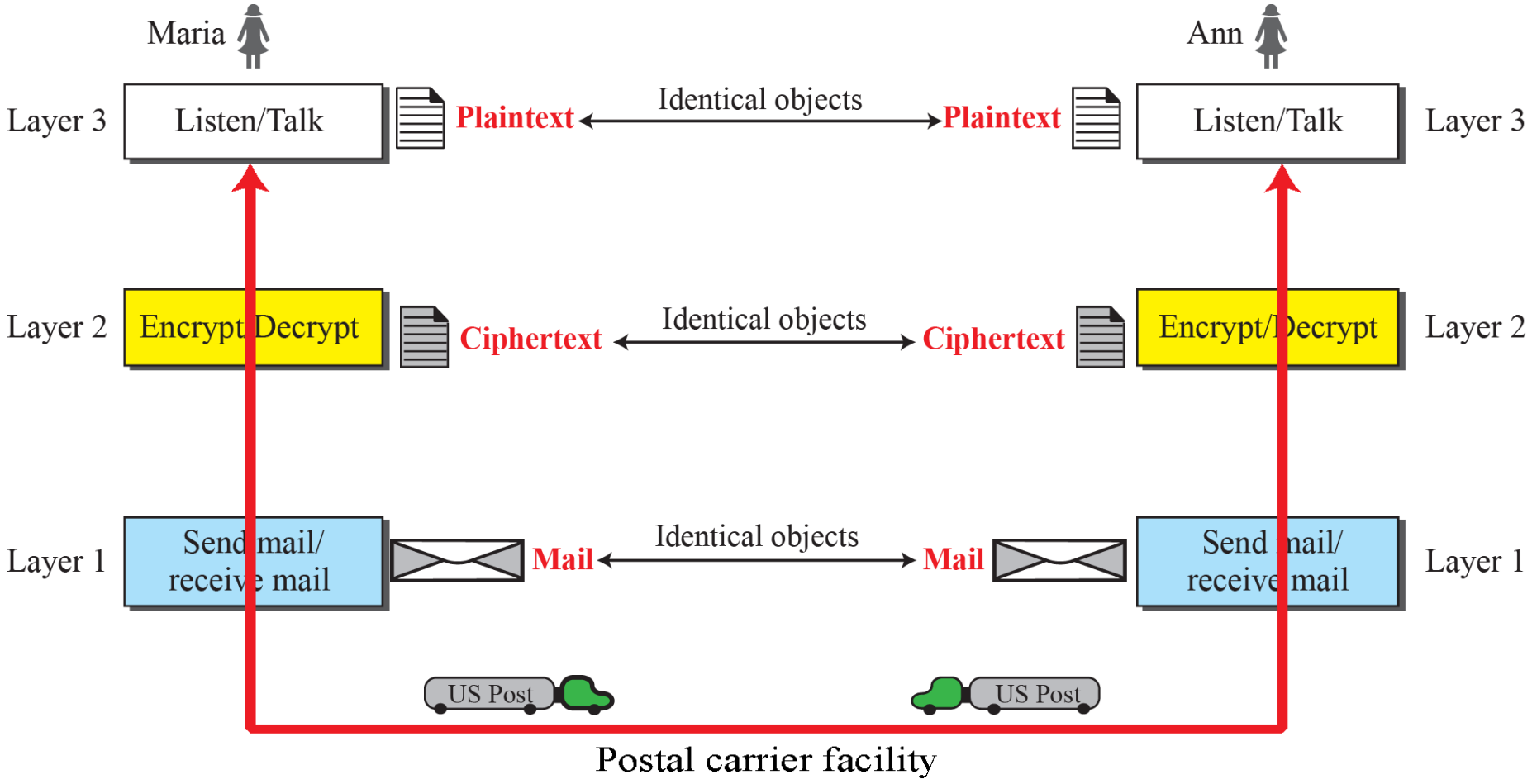
Scenario-2

- They agree on encryption/Decryption technique.(Assume that both have a common technique that makes it hard to decrypt the letter without a key)

- Here, Communication between them takes place in **three layers.**

  - Assume that both have **three machines** (Robots)to perform task at each layer.

A three-layer protocol



Continued…

- Protocol Layering helps us to the divide the task into several smaller & simpler ones.
- A Layer can be defined as a black box with inputs & outputs without concern about how inputs are changed to outputs.
- Protocol Layering allows us to separate the services from implementation.

Continued…

- A Layer should be able to receive set of services from lower layers and give services to the above layers.
- In the Internet, communication does not only use two end systems, intermediate systems also which need only some layers.

- If Protocol Layering is not used, then each intermediate systems will be as complex as end systems and makes the whole system very expensive.

Is single Layer not enough to make job easier.???

Principles of Protocol Layering

Let us discuss two principles of protocol layering.

- The first principle dictates that if we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction.

- The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical.

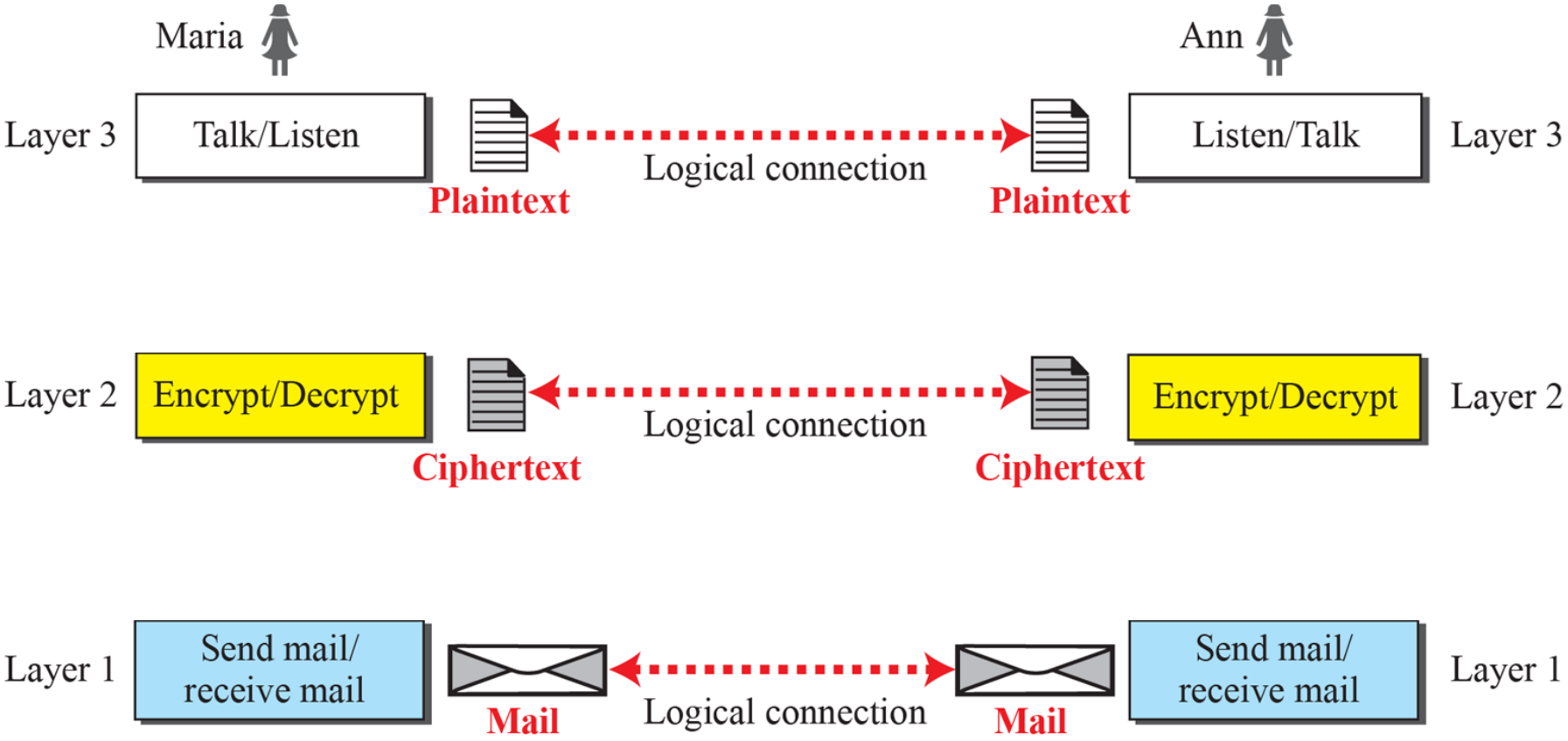
Logical Connections

- After following the above two principles, we can think about logical connection between each layer as shown in Figure, i.e we have **layer-to-layer communication.**

- Maria and Ann can think that there is a logical connection at each layer through which they can send the object created from that layer.

- Discuss the concept of logical connection will help us better understand the task of layering.

Logical connection between peer layers



TCP/IP PROTOCOL SUITE

- TCP/IP is a protocol suite(Set of protocols organized in different layers)used in the Internet today.
- It is a hierarchical protocol made up of interactive modules, each provides specific functionality.
- Hierarchical means that each upper layer protocol is supported by the services provided by the lower layers protocols.
- TCP/IP is a five layer model.

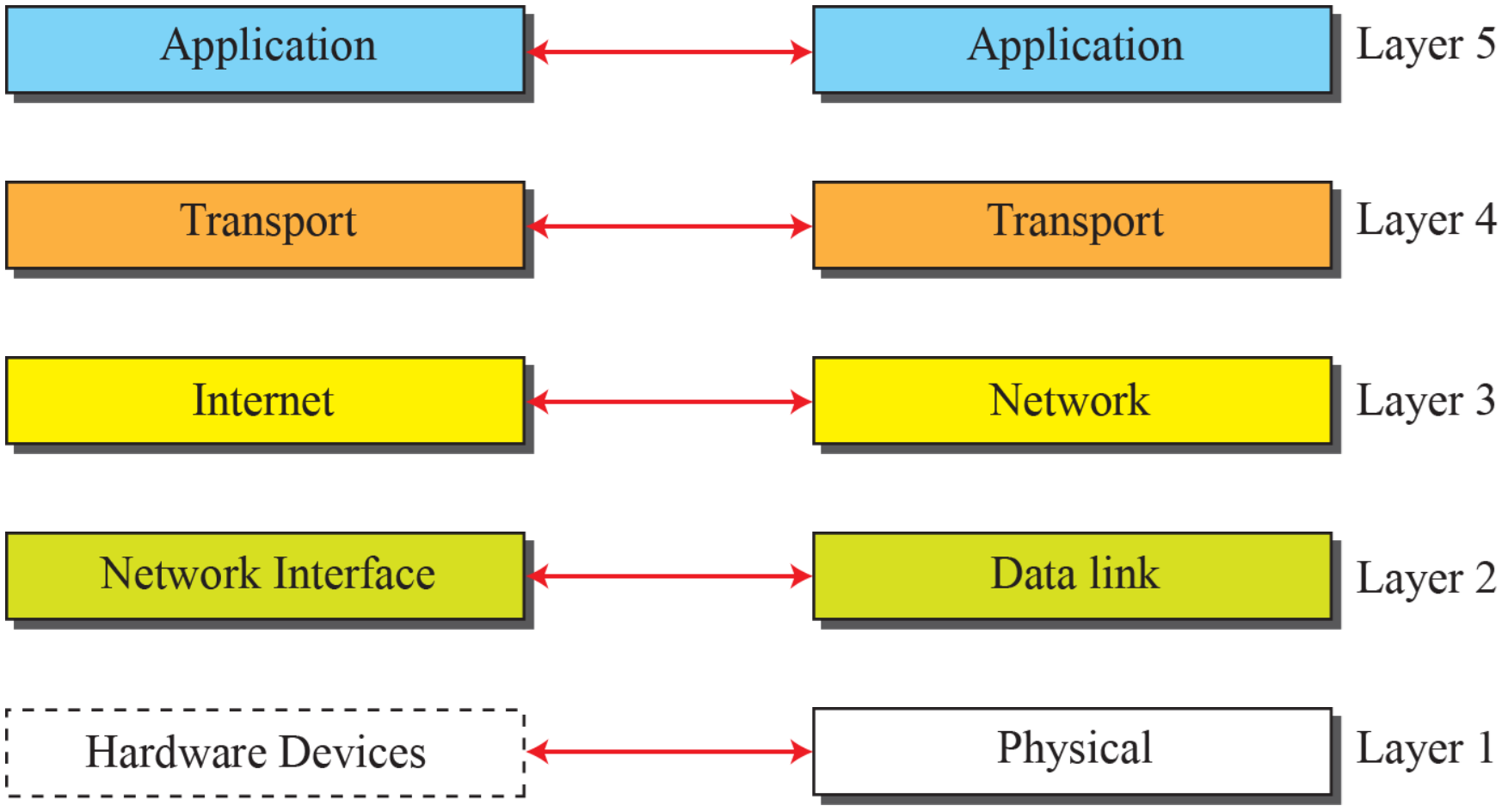
Layered Architecture

- To show how the layers in the TCP/IP protocol suite are involved in communication between two hosts,

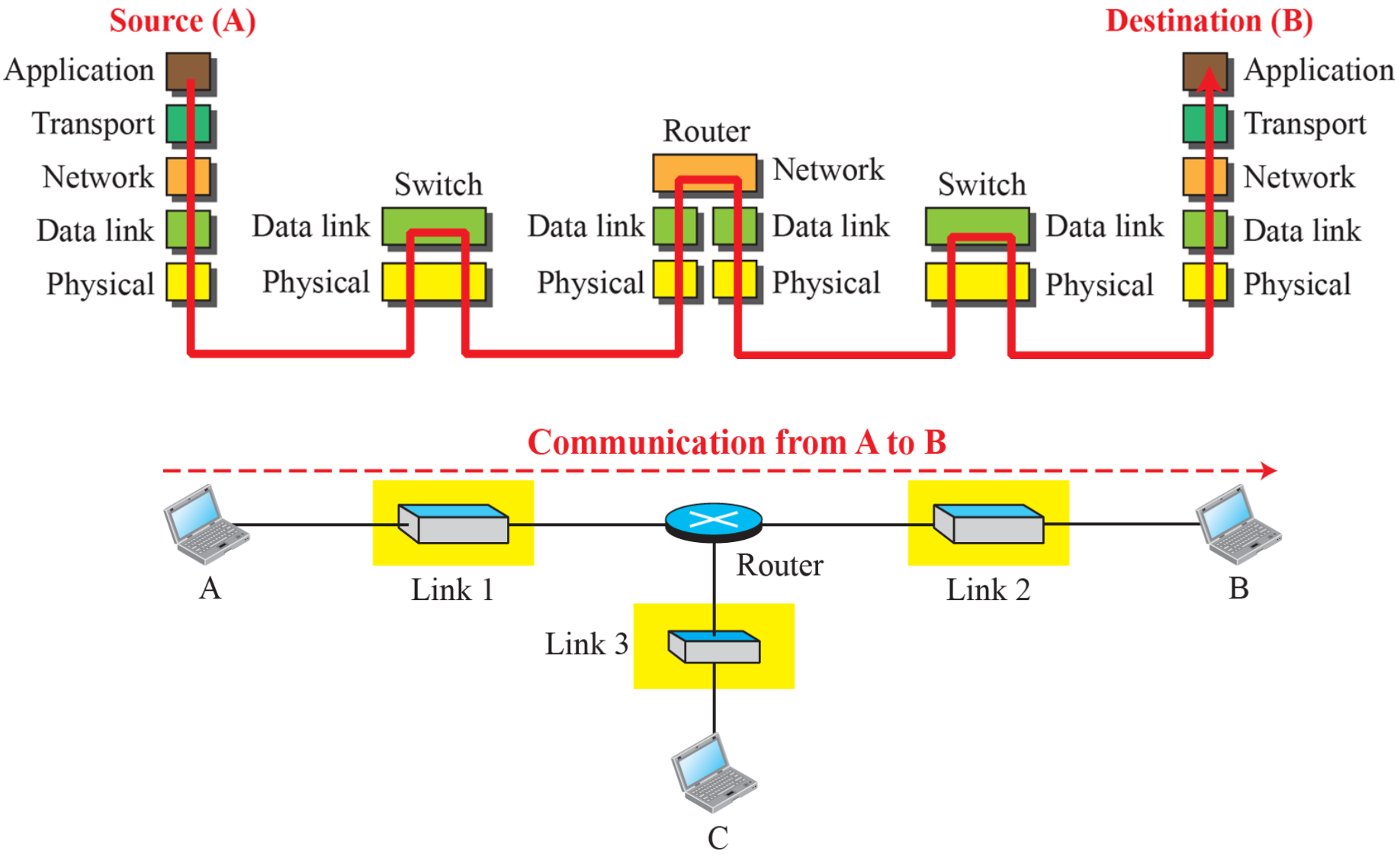
➢ Assume that protocol suite is used in a small internet made up of three LANs (links), each with a link-layer switch.

➢ Assume that the links are connected by one router, as shown in Figure below.

Layers in the TCP/IP protocol suite



Communication through an internet



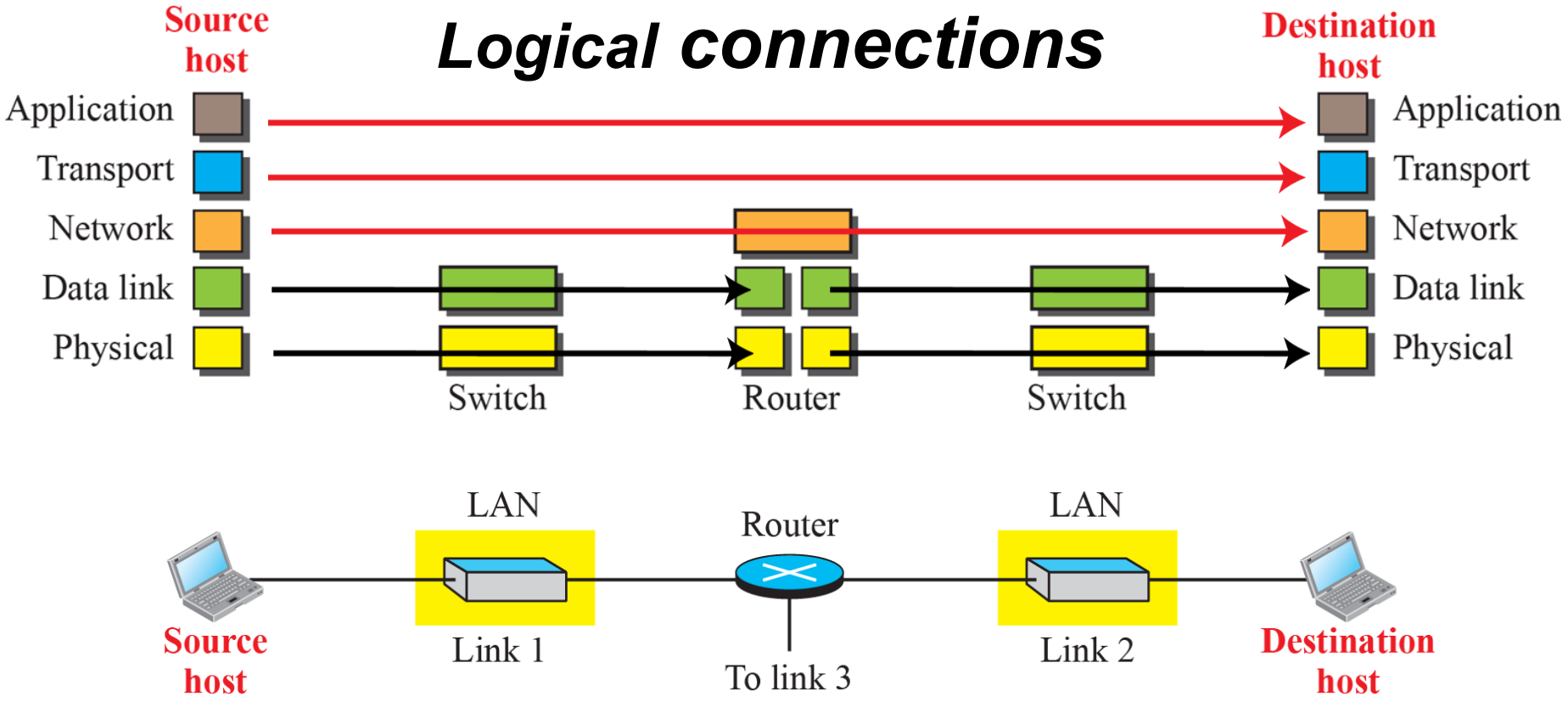
Layers in the TCP/IP Protocol Suite

- Discuss the functions and duties of layers in the TCP/IP protocol suite.

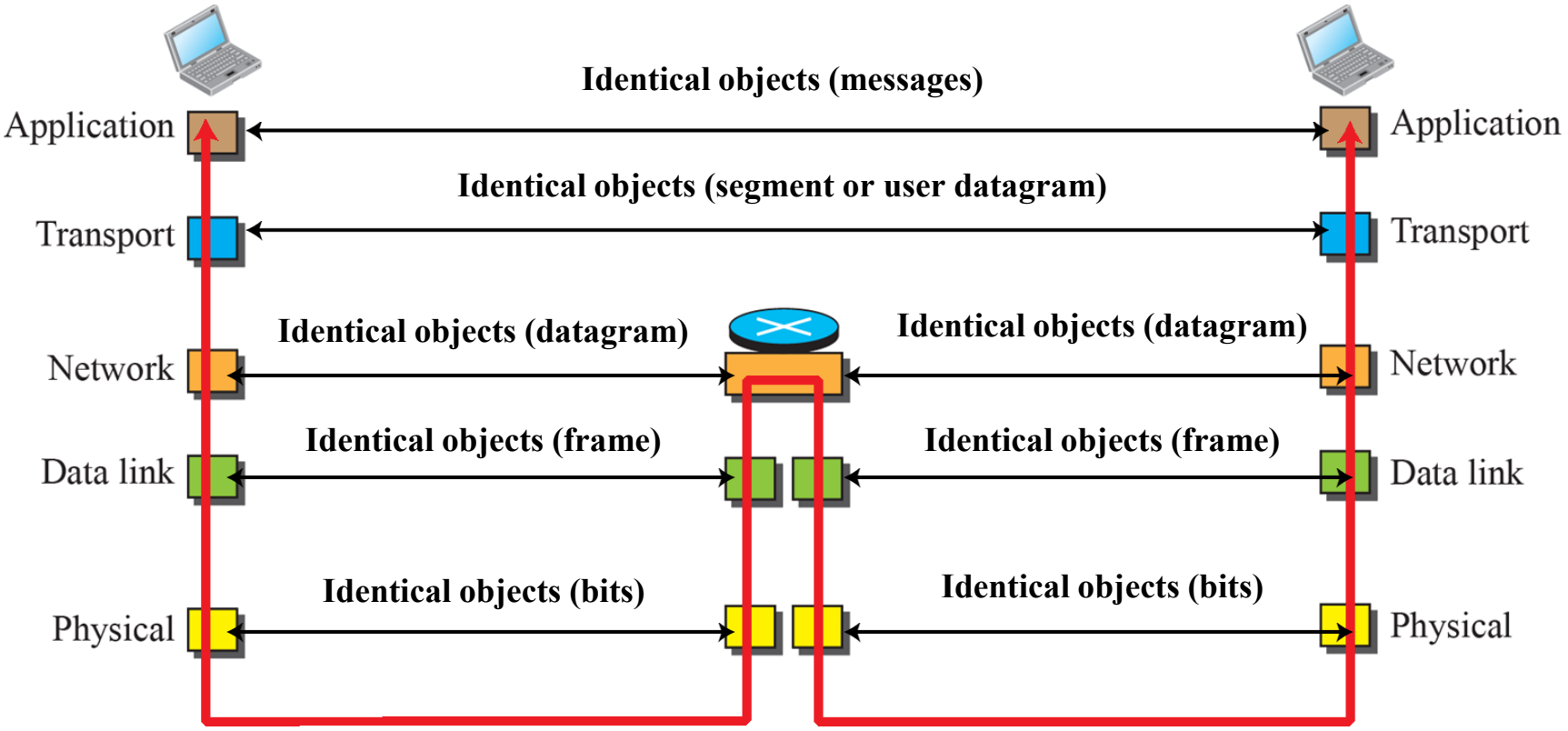
- To better understand the duties of each layer, we need to think about the logical connections between layers in the simple Internet.

- ***Duties of AL,TL & NL is end-to-end.***
- ***Duties of DLL and PL is hop-to-hop.***

Logical connections between layers in TCP/IP



Identical objects in the TCP/IP protocol suite



Description of Each Layer:

1. Physical Layer

- Responsible for

1. *Physical characteristics of interface and medium.*

   Transmitting individual bits from one device to the transmission medium.

2. *Representation of bits.*

Data consists of a stream of bits(0s,1s) with no interpretation. Bits must be encoded into signals- Electrical or optical.

1. Physical Layer

3. *Data rate*

Defines number of bits sent in each second.

4. *Synchronization of bits*

Sender and receiver must use same bit rate i.e sender and receiver clocks must be synchronized

5. *Line configuration*

Concerned with connection of device to a medium

- Point-to-point Two devices are connected with dedicated link.
- Multi point Link is shared among multiple devices.

1. Physical Layer

6. *Physical topology*

   Defines how devices are connected to make network.

   Defines various topologies used in the underlying network.

7. *Transmission mode*

   Defines direction of transmission between devices (Simplex, half duplex or Full duplex)

2. Data Link Layer

- As Internet is made up of several links connected by routers, routers are responsible for choosing best links and DLL is responsible for

1. Taking data grams and moving it across the link. The link can be wired LAN or wireless LAN

2. ***Framing***

Divides stream of bits received from NL into manageable data units called ***frames.***

Continued

3. *Physical addressing*

Attaches the address of device that connects the network to next.

4. *Flow control*

Imposes a flow control mechanism to avoid overwhelming the receiver.

5. *Error Control(ED & EC)*

Adds reliability mechanism to detect and retransmit damaged or lost frames, recognize duplicate frames.

6. *Access Control*

Determine which device has complete control over the link at any given time

3. Network Layer

- Responsible for creating connection between the source and destination computer.

- Source to destination delivery of packet across same/multiple networks.

  if two nodes are connected to same N/W, there is no need for N/W Layer. If, attached to different N/W, N/W Layer is required to accomplish source destination delivery of info.

- Main duty is to provide internetworking, the logical gluing of heterogeneous N/Ws together to look like a single N/W to upper layers.

Functions

1. *Logical addressing*

Attaches logical address of sender and receiver.

2. *Routing*

Uses Connecting devices(Switches/Routers) to switch packet to final destination for choosing best route for each packet.

Network layer

Why Network Layer is required?

Can this routing feature can't be added to TL only?

i) Separation of tasks between different layers.

ii) Routers do not need AL and TL.

Continued..

- The Network Layer in the Internet includes the main protocol IP, that defines format of the packet called a Datagram.

- IP also defines format & structure of the addresses used in this Layer.

- IP is responsible for routing packet from source to its destination, which is achieved by each router forwarding the datagram to the next router in its path.

Continued..

- IP Is the transmission mechanism used by TCP/IP
- *Unreliable and connectionless protocol*
- *P*rovides *Best-Effort Service*
- Best effort means that *IP provides no error checking, does its best to transmits to its destination without any guarantee.*
- If any of these services are required by an application, it should rely on TL protocols.

Continued..

- Network layer also includes Unicast and multicast routing protocols.

- Routing protocols are mainly responsible for forwarding the datagram to the next router in its path.
- Those protocols does not take part in routing.

- N/W Layer also includes auxiliary protocols that help IP in its delivery and routing tasks.

Eg: ICMP, IGMP,DHCP, ARP,RARP

4. Transport Layer

Responsible for

- End-to-End logical connection.
- TL at source hosts gets msg from AL, encapsulates it in TL packet(segment or User Datagram in diff protocol) and sends it through logical connection to TL at the destination host.
- TL gives services to AL.

Transport Layer (Cond..)

- TL has more than one protocol and each AL program can use the protocols that matches his requirements.
- TL has two main protocols:- TCP & UDP
- TCP is connection oriented protocol that first establishes connection between the TL at two hosts before transferring data.
- TCP provides Flow Control, Error Control and Congestion control.

Continued..

- Second protocol is UDP, is connectionless protocol, that transmits the datagrams without creating logical connection.
- In datagram is an independent entity.
- UDP does not provide FC,EC and CC.
- Third Protocol is SCTP(Stream control Transmission Protocol) –is built to respond to emerging multimedia applications.

5. Application Layer

- Communication at AL is between two processes.
- To communicate, a process sends a request to other process and receives the response.
- Many predefined protocols are included.
- User can create a pair of processes to be run at the two hosts.

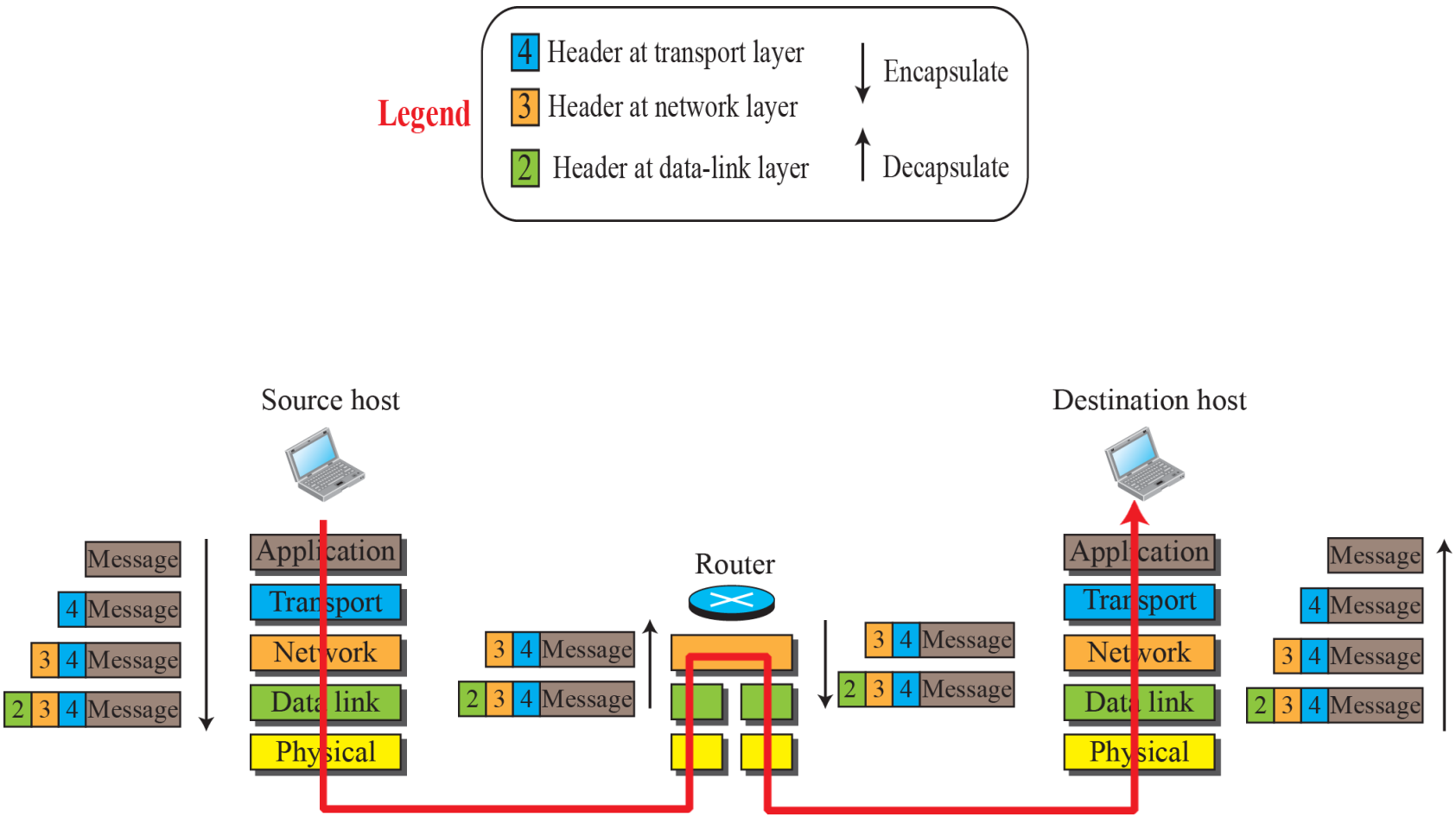
Eg: HTTP, SMTP, FTP, TELNET, SSH, SNMP, DNS, IGMP

Encapsulation and Decapsulation

- One of the important concepts in protocol layering in the Internet is encapsulation/ decapsulation.

- Figure below shows this concept for the small internet.

Encapsulation / Decapsulation



Addressing

- Another concept related to protocol layering in the Internet, addressing.

- Discuss the logical communication between pairs of layers in this model.

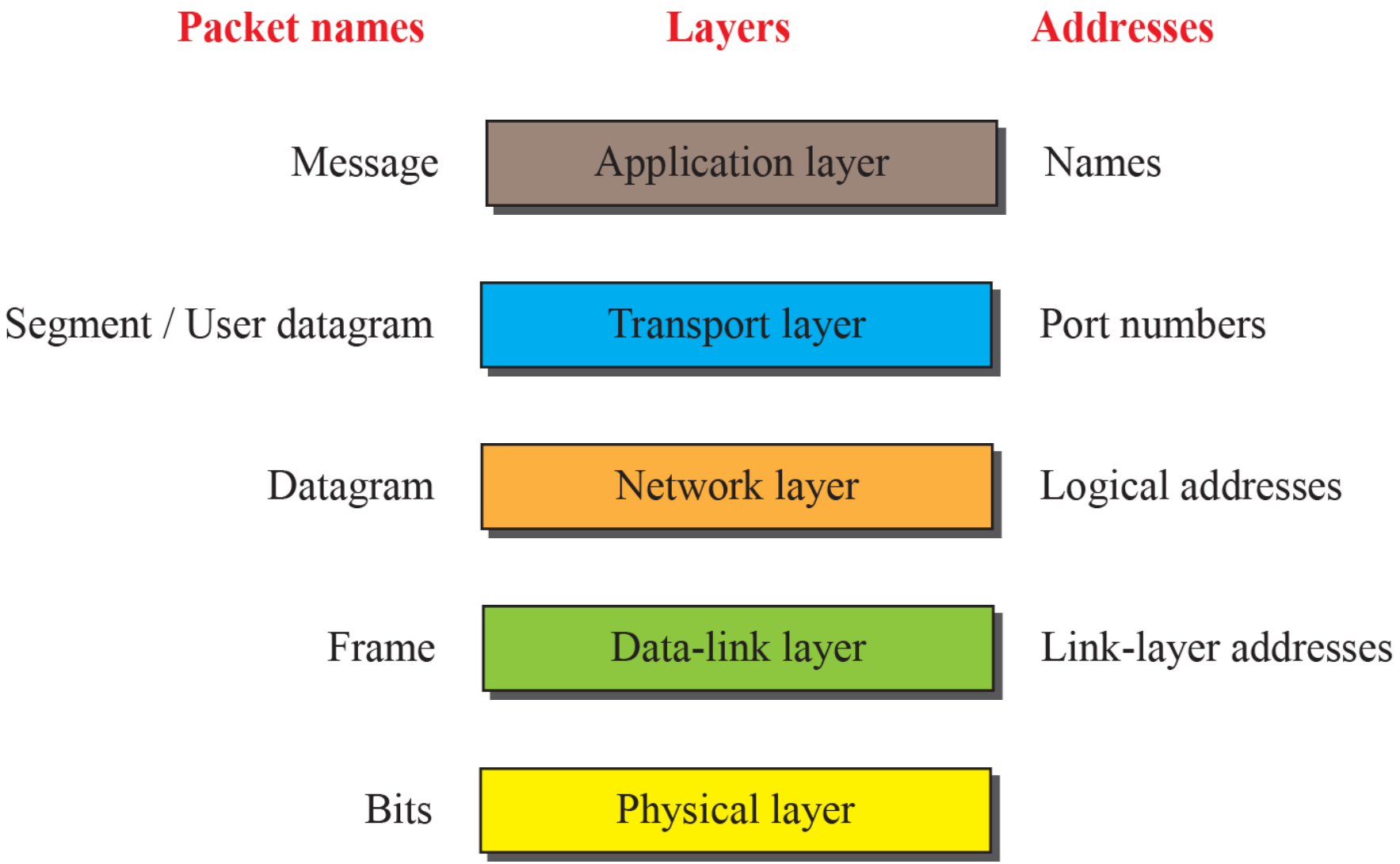
Addressing

- Any communication that involves two parties needs two addresses: source address and destination address.

- Although it looks as if we need five pairs of addresses, one pair per layer, normally have only four because the physical layer does not need addresses;

- The unit of data exchange at the physical layer is a bit, which definitely cannot have an address.

Addressing in the TCP/IP protocol suite



Multiplexing and Demultiplexing

- TCP/IP protocol suite uses several protocols at some layers.

➢ we can say multiplexing at the source and demultiplexing at the destination.

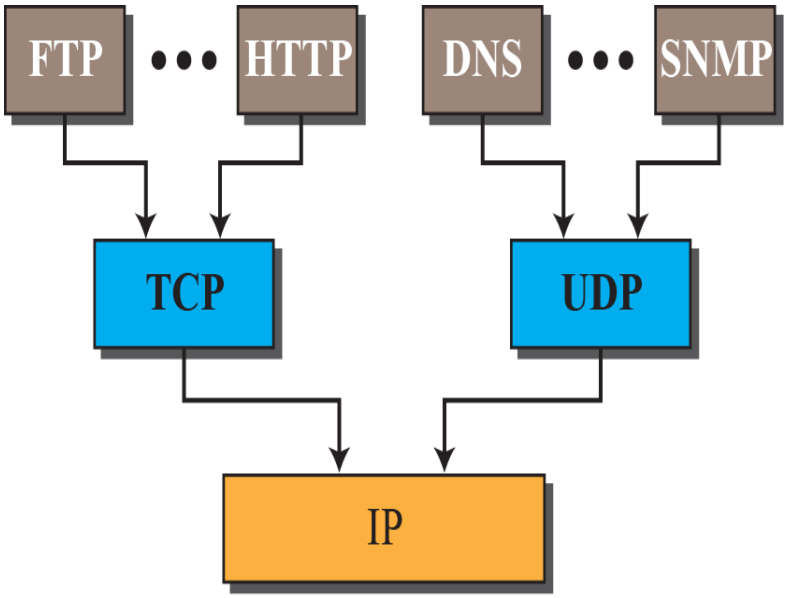
➢ Multiplexing in this case means that a protocol at a layer can encapsulate a packet from several next-higher layer protocols (one at a time);

Multiplexing and Demultiplexing

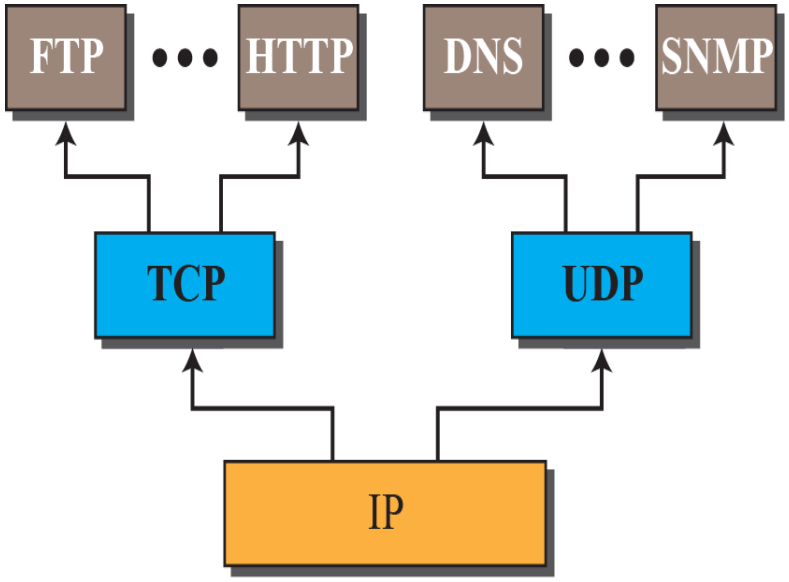
- Demultiplexing means that a protocol can decapsulate and deliver a packet to several next-higher layer protocols (one at a time).

- Figure below shows the concept of multiplexing and demultiplexing at the three upper layers.

Multiplexing and demultiplexing



a. Multiplexing at source



b. Demultiplexing at destination

OSI MODEL

- A protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.

- When communication is simple, we may need only one simple protocol;

- When the communication is complex, we need a protocol at each layer, or protocol layering.

The OSI model

- An ISO(International organizations for standardization) that covers all aspects of network communication is OSI(Open system Interconnection) model introduced in late 1970's

- *Open Systems is a set of protocols that allow any two different systems to communicate regardless of their underlying architecture.*

- Purpose of OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of underlying hardware and software.

The OSI model

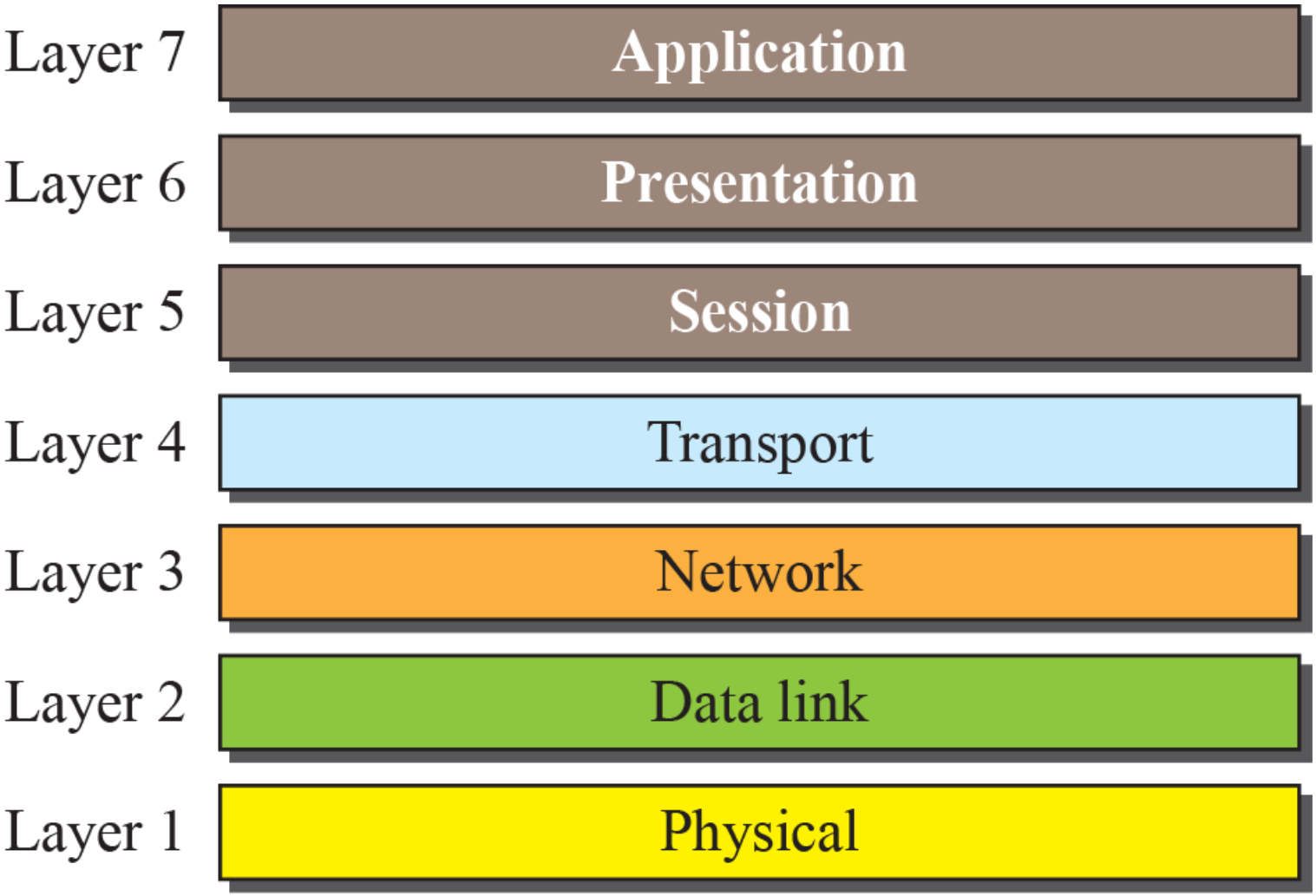
- It is not a protocol.

- It is model for understanding and designing network architecture that is *flexible, robust and interoperable* Is a layered framework for design of network systems that allows communication between all types of computer systems.

The OSI model

- Consists of 7 separate but related layers.
- Each layer *defines a part of process of moving information* across a network

The OSI model



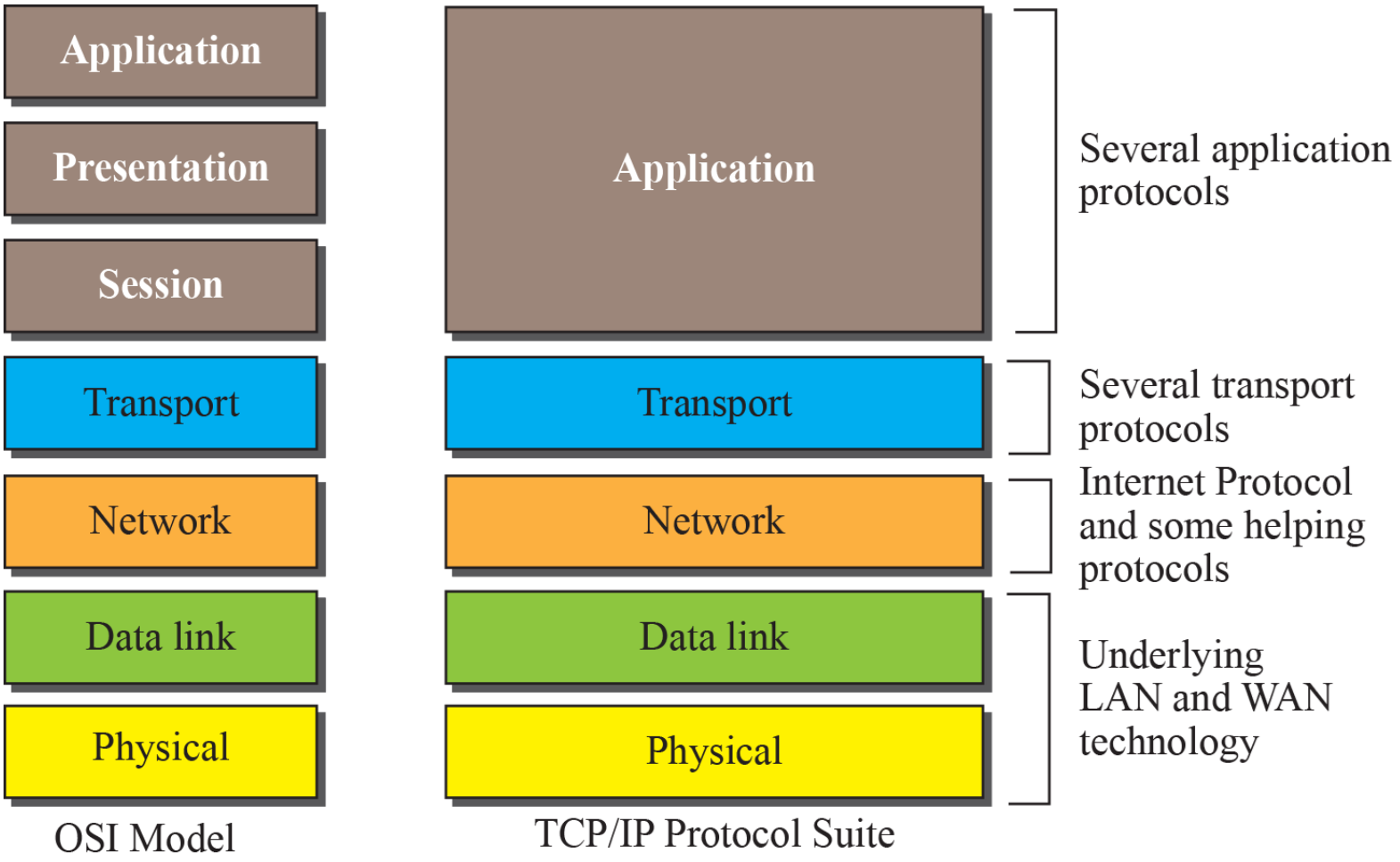
OSI versus TCP/IP

- When we compare the two models, we find that two layers, session and presentation, are missing from the TCP/IP protocol suite.

- These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model.

- The application layer in the suite is usually considered to be the combination of three layers in the OSI model.

TCP/IP and OSI model



Lack of OSI Model's Success

- The OSI model appeared after the TCP/IP protocol suite.

- Most experts were at first excited and thought that the TCP/IP protocol would be fully replaced by the OSI model.

Lack of OSI Model's Success

- This did not happen for several reasons

1. OSI was completed when TCP/IP was fully in place.

1. Some Layers in OSI was never fully defined.(Presentation Layer & Session Layer)

1. OSI was implemented by an organization in a different application (Performance level was low compared to TCP/IP).

Module 3

10.1 INTRODUCTION

10.2 BLOCK CODING

10.3 CYCLIC CODES

10.4 CHECKSUM

Chapter 10: Objective

- The first section introduces types of errors, the concept of redundancy, and distinguishes between error detection and correction.

- The second section discusses block coding. It shows how error can be detected using block coding and also introduces the concept of Hamming distance.

- The third section discusses cyclic codes. It discusses a subset of cyclic code, CRC, that is very common in the data-link layer. Also shows how CRC can be easily implemented in hardware and represented by polynomials.

Chapter 10: Objective (continued)

- The fourth section discusses checksums. It shows how a checksum is calculated for a set of data words. It also gives some other approaches to traditional checksum.

10-1 INTRODUCTION

- TCP/IP protocol does not define any protocol in DLL or PL.
- These two layers are territories of networks that when connected make up the Internet.
- These networks (wired/wireless), provide services to upper three layers of TCP/IP suite.

- Networks must be able to transfer data from one device to another with acceptable accuracy.
- For most of the applications, system must guarantee that data received are identical to the one transmitted .

INTRODUCTION

- In transmission, there are possibilities that any time message gets corrupted.
- Many factors alter one or more bits of a message.
- Some applications require a mechanism for detecting and correcting errors.
- Some applications can tolerate some errors.

Eg: A/V (random errors)

but transfer of text, expect high level of accuracy.

- Discuss some issues related, directly or indirectly, to error detection and correction.

Types of Errors

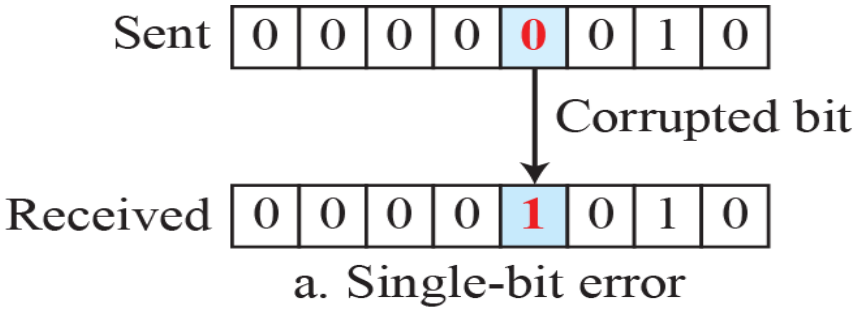
- Whenever bits flow from one point to another, they are subject to unpredictable changes because of interference.

- This interference can change the shape of the signal.

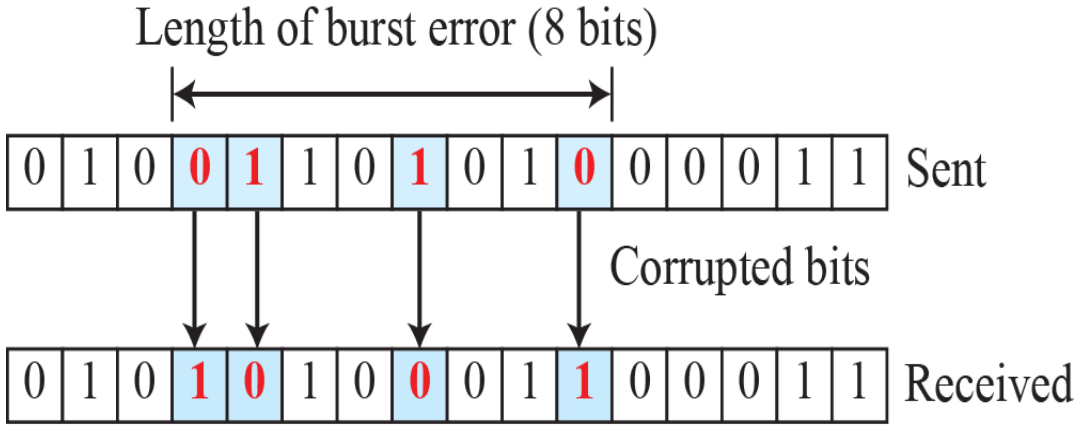
- The term single-bit error means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1.

- The term burst error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1. Figure below shows the effect of a single-bit and a burst error on a data unit.

Single-bit and burst error



a. Single-bit error



b. Burst error

Types of Errors

- Burst error is more likely to occur than single bit error.
- Reason is duration of noise signal is longer than the duration of a bit.
- When noise affects data, it affects set of bits.
- Number of bits affected depends on the duration of the noise and data rate.

Eg: If we send 1 kbps of data, a noise 1/100 s can affect 10 bits.
If 1 Mbps data rate, noise can affect 10,000 bits.

Redundancy

- The central concept in detecting or correcting errors is redundancy.

- To be able to detect or correct errors, need to send some extra bits with our data.

- These redundant bits are added by the sender and removed by the receiver.

- Presence of redundant bits allows the receiver to detect or correct corrupted bits.

Detection versus Correction

- The correction of errors is more difficult than the detection.

- In error detection, check if any error has occurred. The answer is a simple yes or no.Not interested in the number of corrupted bits.

- A single-bit error is the same for us as a burst error.

- In error correction,

  Need to know the exact number of bits that are corrupted and, more importantly, their location in the message.

Coding

- Redundancy is achieved through various coding schemes.

- The sender adds redundant bits through a process that creates a relationship between the redundant bits and the actual data bits.

- The receiver checks the relationships between the two sets of bits to detect errors.

- The ratio of redundant bits to data bits and the robustness of the process are important factors in any coding scheme.

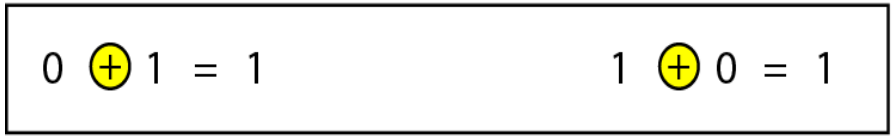
Coding

- Two coding schemes are

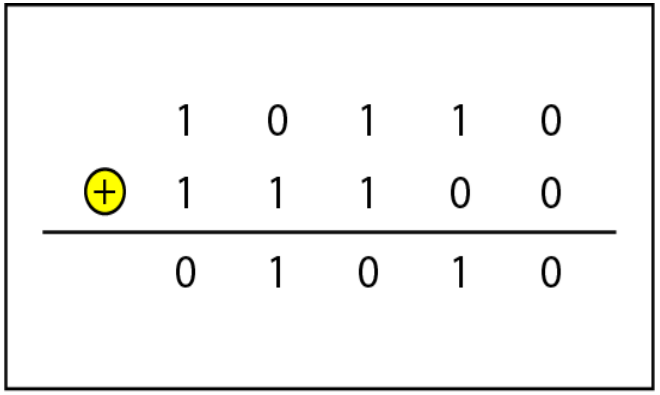
  i) Block coding ii) Convolution coding
- Concentrate on block codes.
- In modulo-N arithmetic, the integers in the range 0 to N −1 inclusive is used,



a. Two bits are the same, the result is 0.



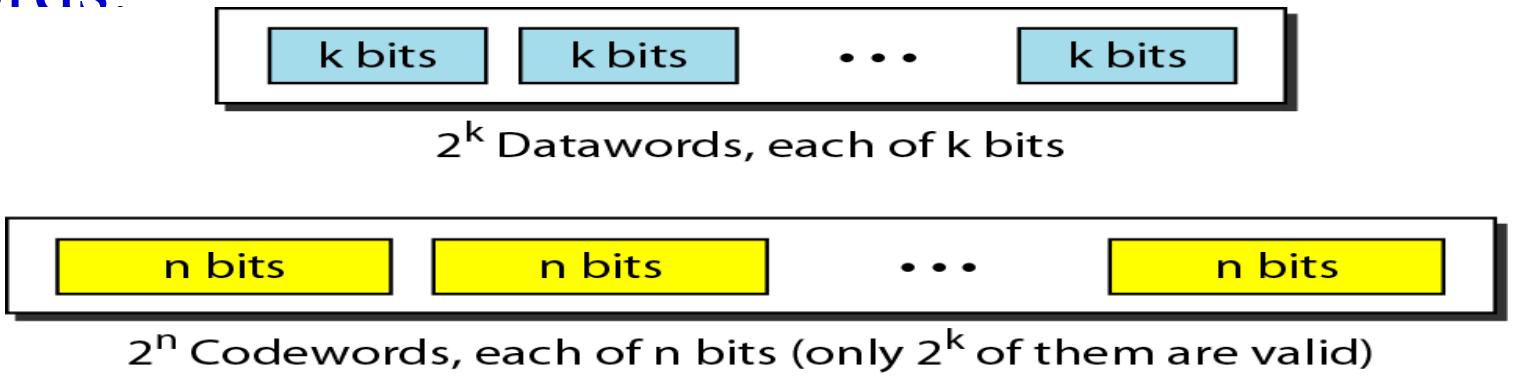
b. Two bits are different, the result is 1.



c. Result of XORing two patterns

10-2 BLOCK CODING

- In block coding, message is divided into blocks, each of k bits, called data words.
- Add r redundant bits to each block to make the length $n = k + r$.
- The resulting n-bit blocks are called code words.



BLOCK CODING

- In block coding, we have set of data words, each of size k, code words, each of size n.
- Since n>k, the possible number of code words are larger than the data words.
- Block coding process is one-to-one, same data word is encoded as code word.
- $2^n - 2^K$ code words are not used.
- If receiver receives invalid codeword, it indicates data is corrupted during transmission.

Error Detection

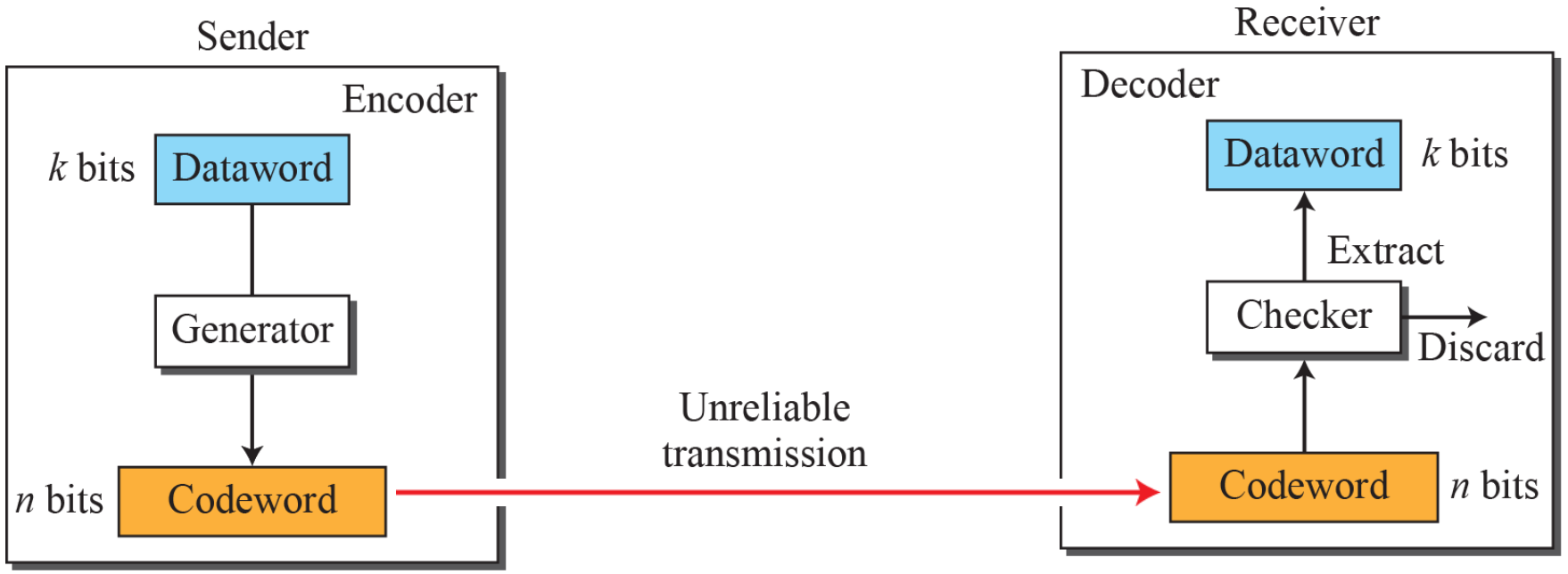
- How can errors be detected by using block coding?

- If the following two conditions are met, the receiver can detect a change in the original codeword.

a. The receiver has (or can find) a list of valid code words.

b. The original codeword has changed to an invalid one.

Process of error detection in block coding



Example 10.1

Let us assume that k = 2 and n = 3. Table below shows the list of data words and code words.

A code for error detection

Datawords	*Codewords*	*Datawords*	*Codewords*
00	000	10	101
01	011	11	110

Assume the sender encodes the dataword 01 as 011 and sends it to the receiver. Consider the following cases:

1. The receiver receives 011. It is a valid codeword. The receiver extracts the dataword 01 from it.
2. The codeword is corrupted during transmission, and 111 is received (the leftmost bit is corrupted). This is not a valid codeword and is discarded.
3. The codeword is corrupted during transmission, and 000 is received (the right two bits are corrupted). This is a valid codeword. The receiver incorrectly extracts the dataword 00. Two corrupted bits have made the error undetectable.

Hamming distance (HD)

- The Hamming distance between two words(of same size) is the number of differences between corresponding bits. *(total number of 1's).*
- *Why HD in Error detection????*
- *HD between the sent code word & received codeword is the number of bits that got corrupted during transmission.*

Eg: Let us find the Hamming distance between two pairs of words.

$000 \oplus 011$ is 011 (two 1s)

Hamming Distance

- The Hamming distance d(000, 011) is 2.
- The Hamming distance d(10101, 11110) is 3 because

$$10101 \oplus 11110 \text{ is } 01011 \text{ (three 1s)}$$

- *If the HD between sent & received code word is not zero, then code word has been corrupted during transmission.*
- *The minimum Hamming distance is the smallest Hamming distance between all possible pairs in a set of words.*

Hamming distance

- Find the minimum Hamming distance of the coding scheme in

Datawords	*Codewords*
00	000
01	011
10	101
11	110

We first find all Hamming distances

$$d(000, 011) = 2 \quad d(000, 101) = 2 \quad d(000, 110) = 2 \quad d(011, 101) = 2$$
$$d(011, 110) = 2 \quad d(101, 110) = 2$$

The d_{min} in this case is 2

Example 10.2

Let us find the Hamming distance between two pairs of words.

1. The Hamming distance $d(000, 011)$ is 2 because $(000 \oplus 011)$ is 011 (two 1s).
2. The Hamming distance $d(10101, 11110)$ is 3 because $(10101 \oplus 11110)$ is 01011 (three 1s).

Example 10.3

The minimum Hamming distance for our first code scheme (Table 10.1) is 2.

This code guarantees detection of only a single error. For example, if the third code word (101) is sent and one error occurs, the received code word does not match any valid code word.

If two errors occur, however, the received code word may match a valid code word and the errors are not detected.

Example 10.6

In our first code (Table 10.1), the numbers of 1s in the nonzero codewords are 2, 2, and 2. So the minimum Hamming distance is $\boldsymbol{d}_{\text{min}} = 2$.

LINEAR BLOCK CODES

- Almost all block codes used today belong to a subset called *linear block codes*.
- A linear block code is a code *in which the* exclusive OR (addition modulo-2) of two valid code words creates another valid codeword.

Table： Simple parity-check code C(5, 4)

Datawords	Codewords	*Datawords*	Codewords
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

Example

- Let us see if the two codes defined in Tables above, belong to the class of linear block codes.

1. The scheme in the above Table is a linear block code because the result of XORing any codeword with any other codeword is a valid codeword.

   So the minimum Hamming distance is $\boldsymbol{d_{min} = 2}$.

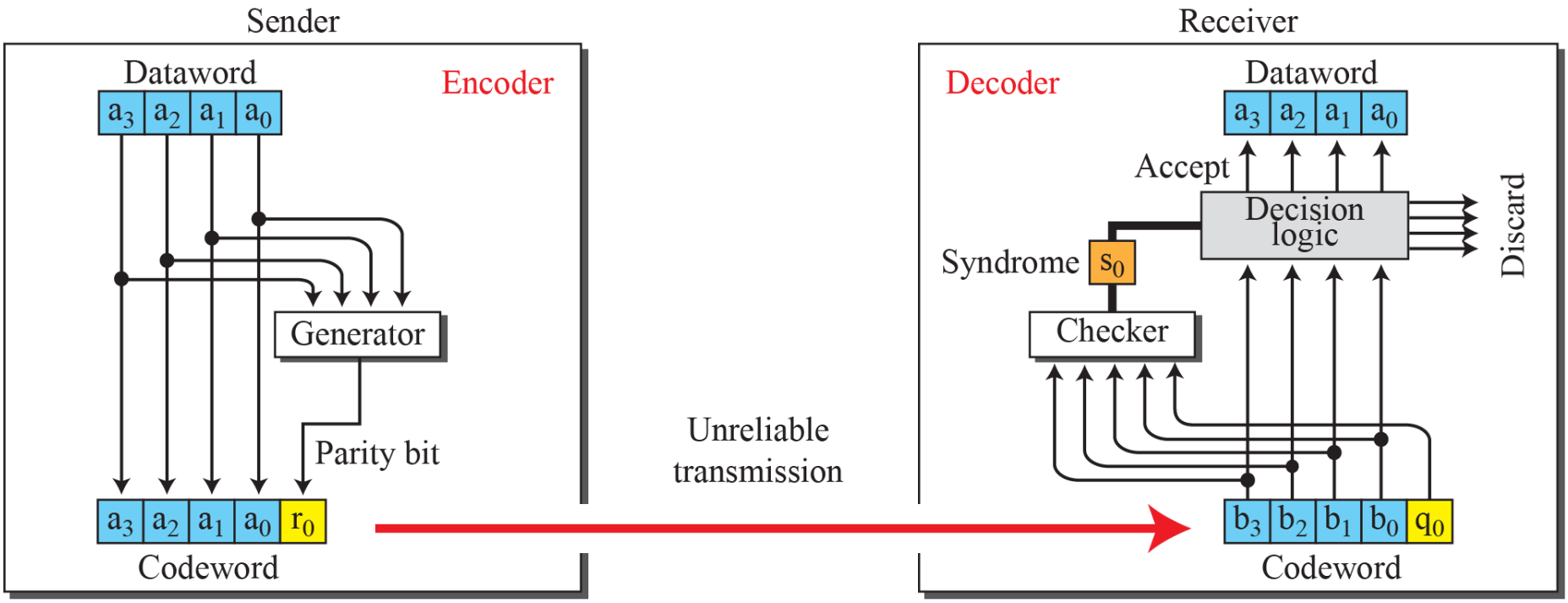
2. The scheme in Table second is also a linear block code. We can create all four codewords by XORing two other codewords.

- *So the minimum hamming distance is* $d_{min} = 3$.

Simple Parity check code

- A simple parity-check code is a single-bit error-detecting code in which $n = k + 1$ with $d_{min} = 2$.
- The extra bit called parity bit is selected to make number of 1's in the code word even.
- The code is single-bit error detecting code.
- It cannot correct any error.
- In the fig. encoder takes a copy of 4 bit data word(a_0, a_1, a_2, a_3) and generates a parity bit r_0.
- Parity bit creates the 5-bit code word.

Encoder and decoder for simple parity-check code



Simple Parity check code

- The parity bit that is added makes the number of 1's in the codeword even. i.e

$$r_0 = a_0 + a_1 + a_2 + a_3 \quad \text{(modulo-2)}$$

- If number of 1s is even, the result is 0.
- If number of 1s is odd, the result is 1.
- The sender sends the code word which is corrupted during transmission.
- The receiver receives 5 bit code word.
- The checker performs modulo-2 of all the bits received. The result is called syndrome.

$$s_0 = b_0 + b_1 + b_2 + b_3 + q_0 \quad \text{(modulo-2)}$$

Continued..

- If number of 1s in the codeword is even, the syndrome is 0.
- If number of 1s in the codeword is odd, the syndrome is 1.
- The syndrome is passed to decision logic analyzer. If Syndrome is 0, no error in the code word received, data portion is extracted.
- If syndrome is 1, error in the received codeword and discarded.

Example 10.7

Let us look at some transmission scenarios. Assume the sender sends the data word 1011. The codeword created from this data word is 10111, which is sent to the receiver. Five cases can be examined:

1. No error occurs; the received codeword is 10111. The syndrome is 0. The dataword 1011 is created.
2. One single-bit error changes a_1. The received codeword is 10**0**11. The syndrome is 1. No dataword is created.
3. One single-bit error changes r_0. The received codeword is 1011**0**. The syndrome is 1. No dataword is created. Note that although none of the dataword bits are corrupted, no dataword is created because the code is not sophisticated enough to show the position of the corrupted bit.
4. An error changes r_0 and a second error changes a_3. The received codeword is **0**011**0**. The syndrome is 0. The dataword 0011 is created at the receiver. Note that here the dataword is wrongly created due to the syndrome value. The simple parity-check decoder cannot detect an even number of errors. The errors cancel each other out and give the syndrome a value of 0.
5. Three bits—a_3, a_2, and a_1—are changed by errors. The received codeword is **010**11. The syndrome is 1. The dataword is not created. This shows that the simple parity check, guaranteed to detect one single error, can also find any odd number of errors.

10-3 CYCLIC CODES

- Cyclic codes are special linear block codes with one extra property.
- In a cyclic code, if a codeword is cyclically shifted (rotated), the result is another codeword.

Eg:1011000 is a codeword and if it is cyclically left-shifted, then 0110001 is also a codeword.

We can shift the bits using

$$b_1=a_0, b_2=a_1, b_3=a_2, b_4=a_3, b_5=a_4, b_6=a_5, b_0=a_6$$

Cyclic Redundancy Check

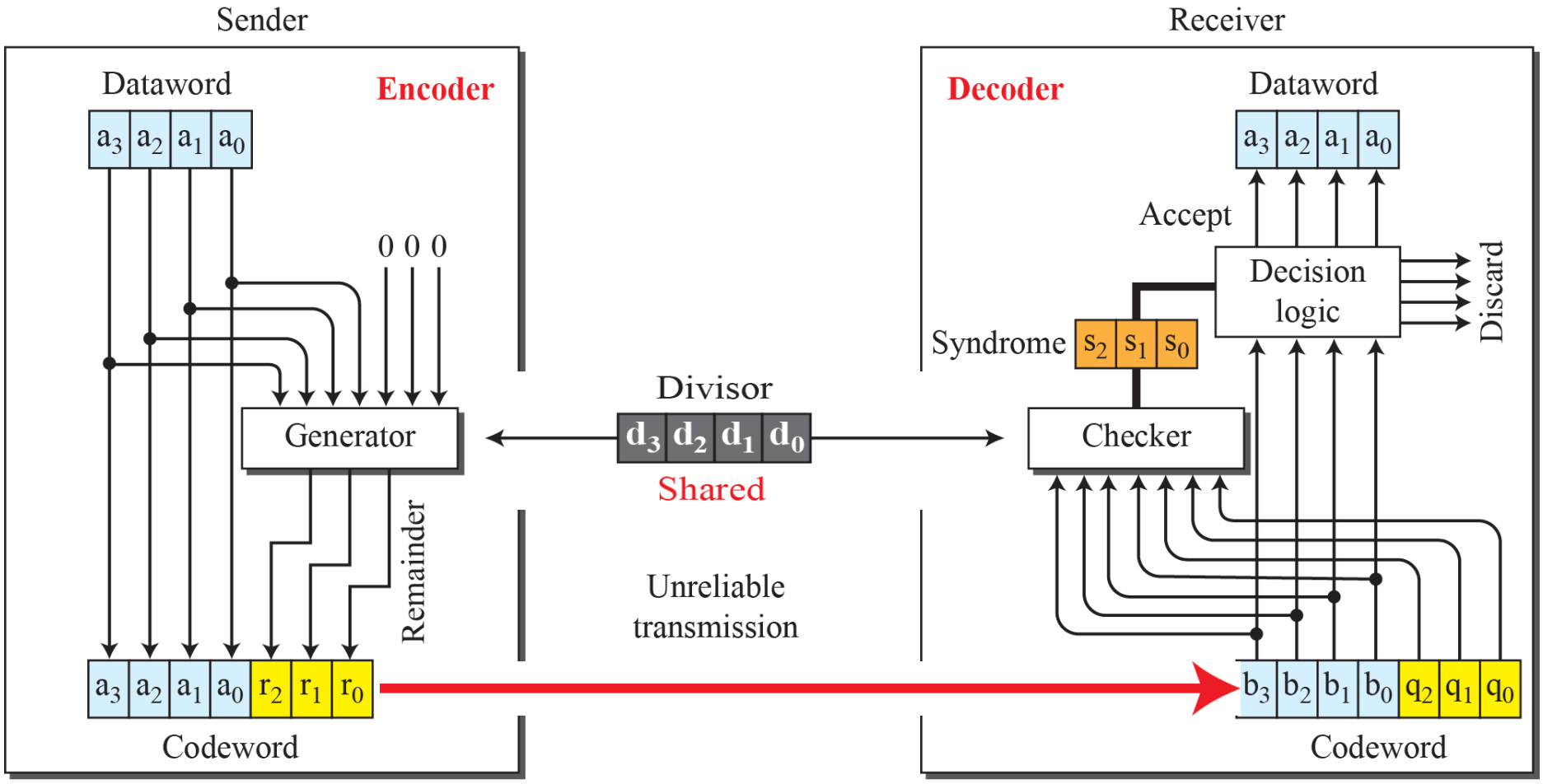
- Cyclic codes are created to correct errors.

- Discuss a subset of cyclic codes called the cyclic redundancy check (CRC), which is used in networks such as LANs and WANs.

A CRC code with C(7, 4)

Dataword	Codeword	Dataword	Codeword
0000	0000000	1000	1000101
0001	0001011	1001	1001110
0010	0010110	1010	1010011
0011	0011101	1011	1011000
0100	0100111	1100	1100010
0101	0101100	1101	1101001
0110	0110001	1110	1110100
0111	0111010	1111	1111111

CRC encoder and decoder



CRC

- In the encoder, the data word has k bits(4 bits), code word has n bits(7 here).
- The data word is augmented by adding n-k 0s to the right hand side of the word.
- The n-bit result is fed to the generator.
- The generator uses a divisor of size

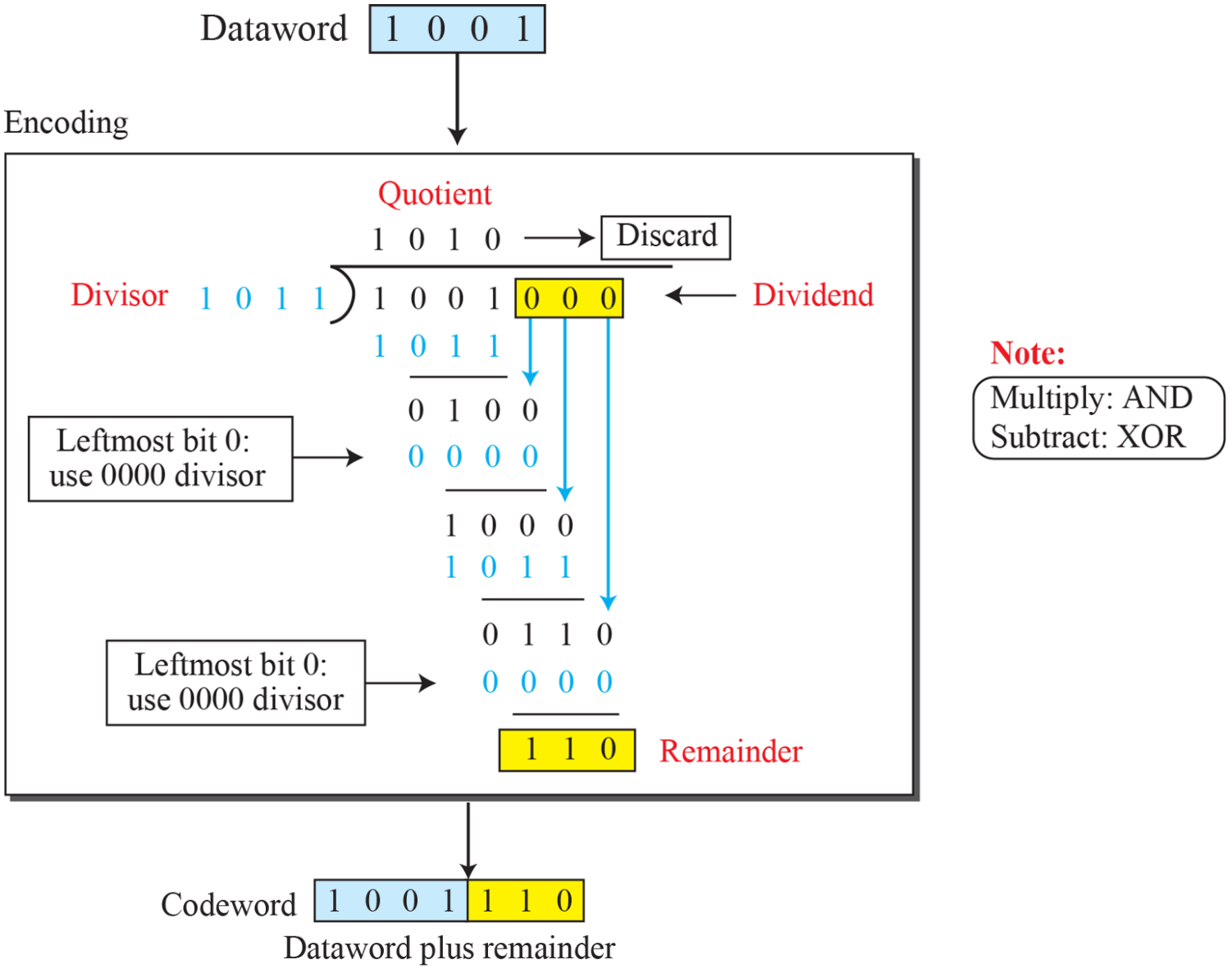
  n-k+1which is predefined and agreed upon.
- The generator divides the augmented dataword

CRC

by divisor (modulo-2 division).

- The quotient of the division is discarded.
- The remainder $r_2 r_1 r_0$ is appended to the data word to create the codeword.
- The decoder receives the corrupted codeword.
- A copy of all n bits are fed to the checker.
- The remainder produced by checker is a syndrome of n-k bits, which is fed to the decision logic analyzer.
- If syndrome bits are 0, the 4 left most bits of the code word are accepted as a data word.

Division in CRC encoder



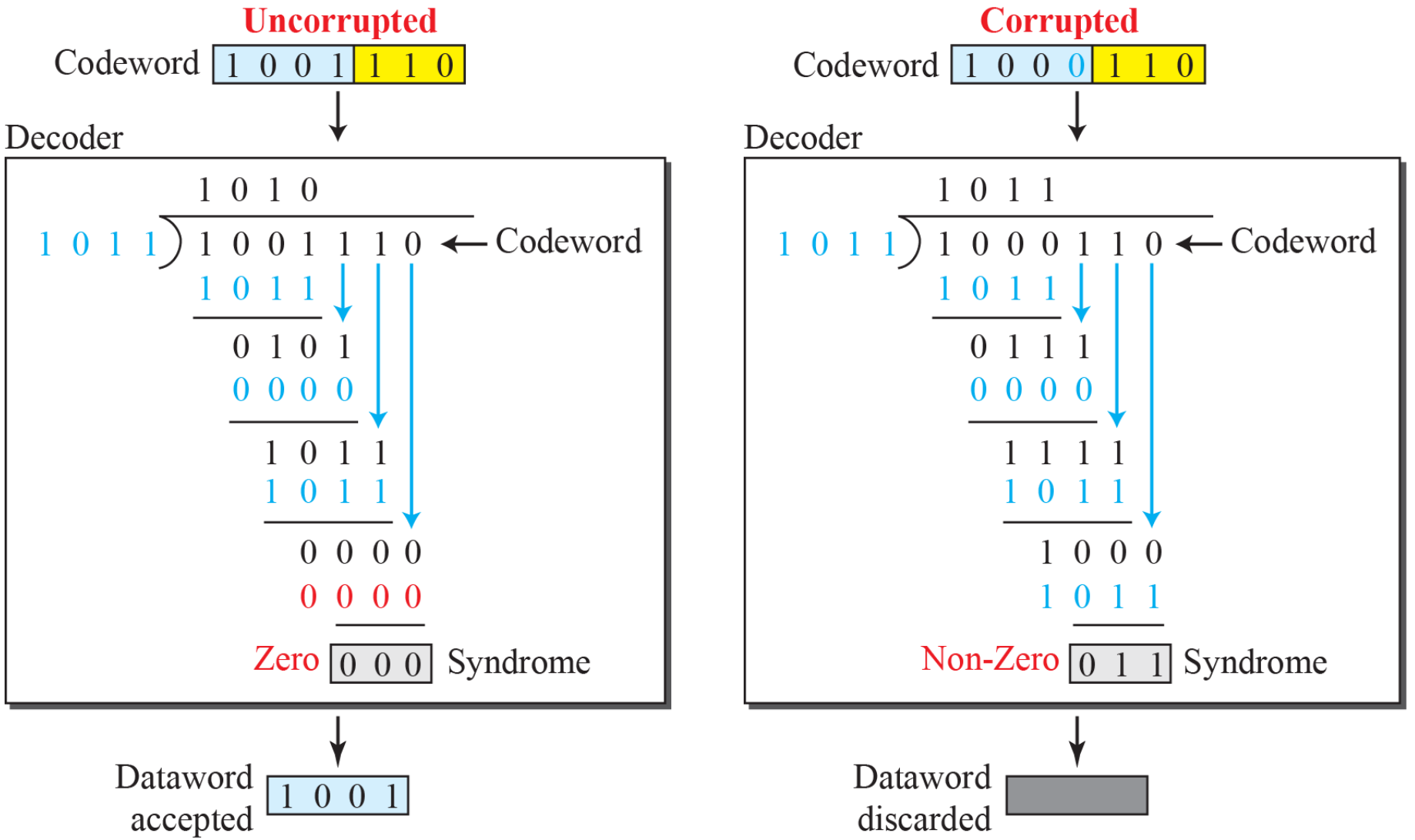
CRC-Encoder

- Encoder takes data word and augments it with n-k number of 0s.
- It then divides the augmented data word by a divisor.
- Modulo-2 binary division is used. For Addition and subtraction, XOR operations is used.
- In each step, copy of the divisor is XOR ed with 4 bits of dividend. The result of XOR operation is 3 bits.

CRC-Decoder

- Code word can change during transmission.
- The decoder same thing as encoder and remainder of the division id the syndrome.
- If syndrome is all 0s, no error, data word is separated from the code word and accepted.
- Otherwise , everything is discarded.
- Question is how divisor 1011 is chosen??

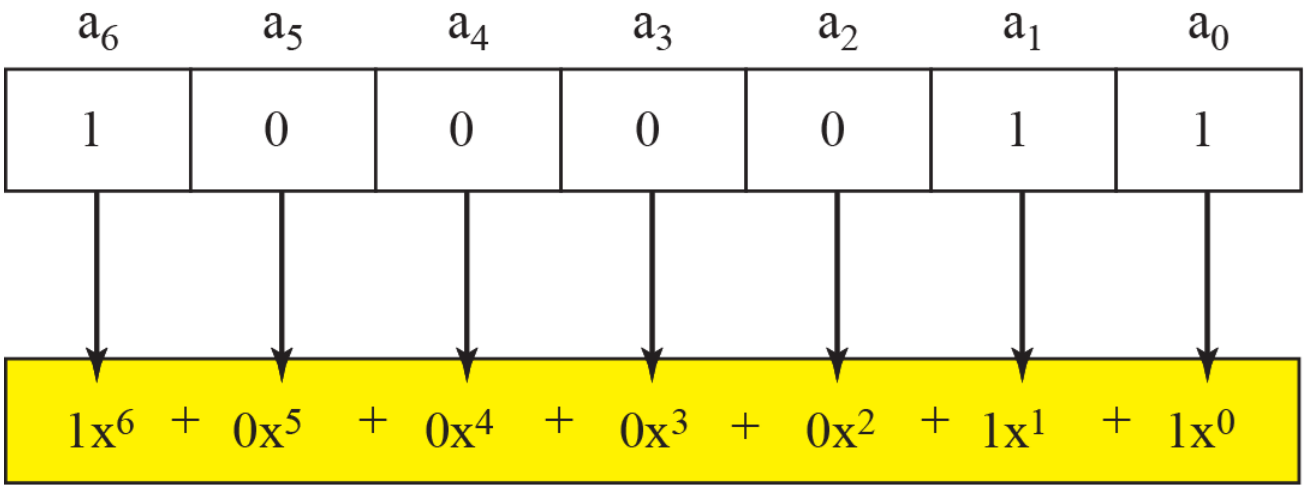
Division in the CRC decoder for two cases



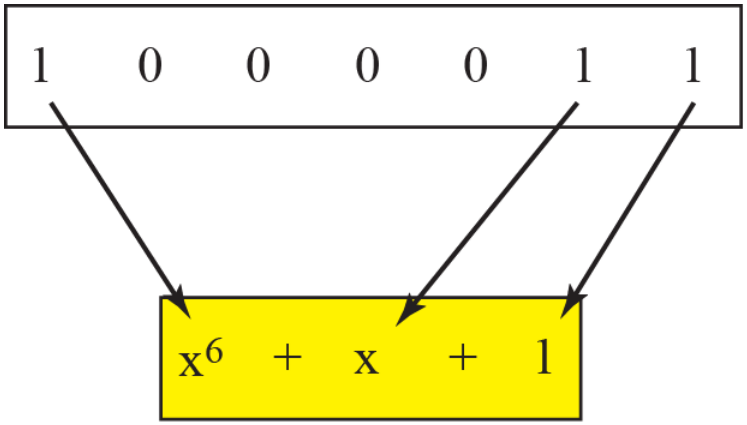
Polynomials

- A better way to understand cyclic codes, and how they can be analyzed is to represent them as polynomials.
- A pattern of 0s and 1s can be represented as polynomial with coefficient of 0 and 1.
- The power of each term shows the position of the bit.
- The coefficient shows the value of the bit.

A polynomial to represent a binary word



a. Binary pattern and polynomial



b. Short form

Degree of Polynomials

- Is the highest power of the polynomial.

Eg: Degree of $x^6 + x+1$ is 6.

- Degree of polynomial is one less than that the number of bits in the pattern.
- Addition and subtraction of polynomials is done by adding and subtracting the coefficient terms with the same power.
- Coefficients are only 0 and 1, and adding is in modulo-2

Adding and subtracting of polynomials

- Addition and subtraction is done by combining terms and deleting pair of identical terms.
- Eg: $x^5 + x^4 + x^2$

$$x^6 + x^4 + x^2$$

$$x^6 + x^5$$

Multiplication and Division of polynomials

- Multiplication is adding powers.

Eg: $x^4 * x^2 = x^6$

- Division, subtract the power of second term from the first term.

Eg: $x^5 / x^2 = x^3$

- Multiplication of two polynomials

$(x^5 + x^3 + x^2)(x^2 + x + 1) = x^7 + x^6 + x^5 + x^5 + x^4 + x^3 + x^4 + x^3 + x^2$

$\Rightarrow x^7 + x^6 + x^2$

Shifting of polynomials

- A binary pattern is shifted left or right by number of bits.
- Shifting to the left means adding extra 0s as right most bits.
- Shifting to the right means deleting some right most bits.
- Shifting to the left is accomplished by multiplying each term of the polynomial by x^m where m is the number of shifted bits

Continued..

- Shifting to the right is accomplished by dividing each term of the polynomial by x^m where m is the number of shifted bits.

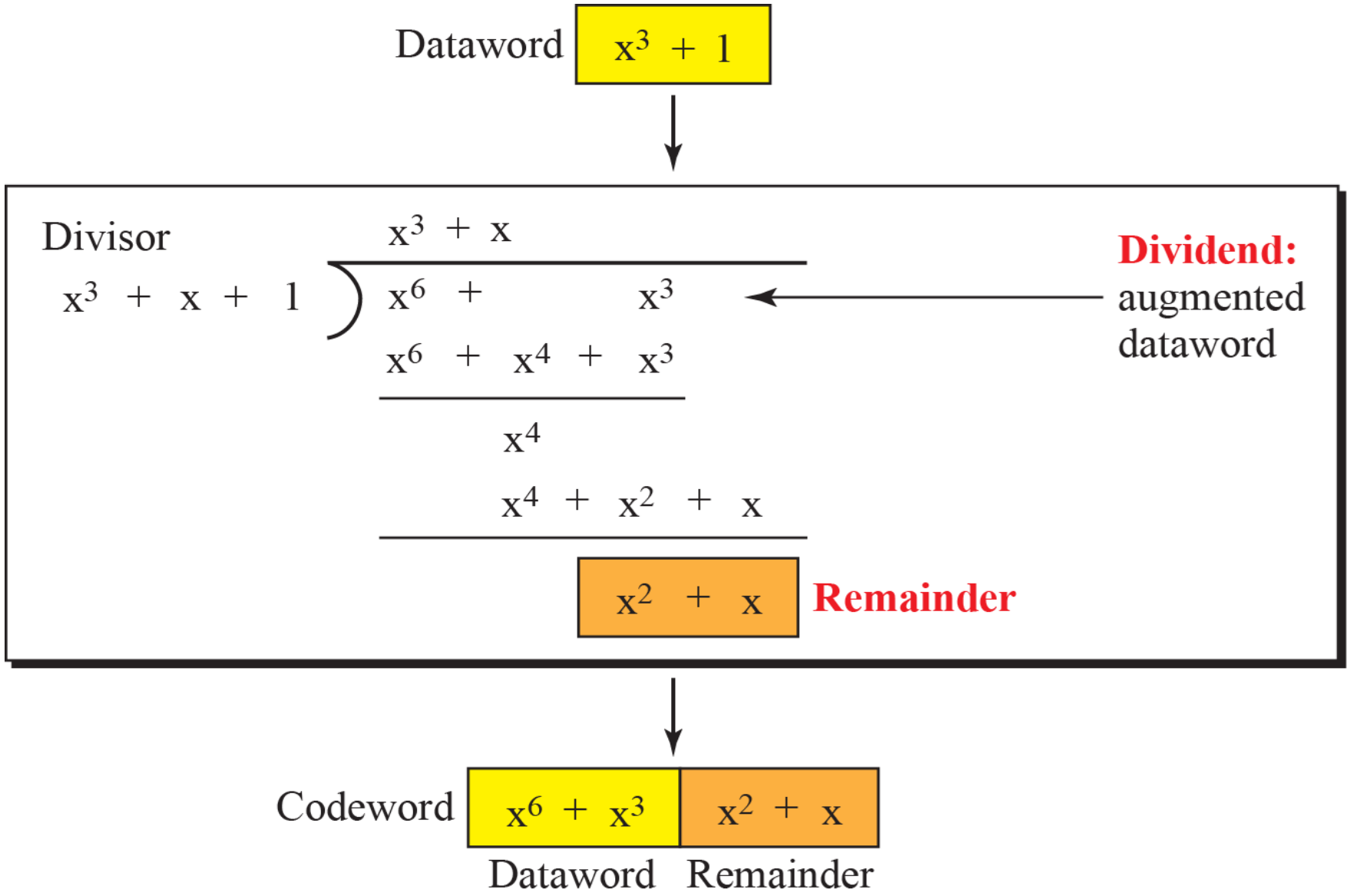
- Shift left by 3 bits :10011 becomes 10011000

  x^4+x+1 becomes $x^7 + x^4+x^3$

- Shift right by 3 bits: 10011 becomes 10

  x^4+x+1 becomes x

CRC division using polynomials



Cyclic code analysis using polynomials

- In a polynomial representation, divisor is normally referred to as generator polynomial t(x).
- f(x) is the polynomial with binary coefficients data word d(x), code word c(x), Generator g(x) syndrome s(x) and error e(x).
  - Is s(x) is not zero, one or more bits are corrupted. If s(x) is zero, either no error or decoder failed to detect any error.

Cyclic Code Analysis

- Analyze a cyclic code to find its capabilities by using polynomials.

- Define the following, where f(x) is a polynomial with binary coefficients.

Dataword: $d(x)$ **Codeword:** $c(x)$

Generator: $g(x)$ **Syndrome:** $s(x)$ **Error:** $e(x)$

Example 10.8

Which of the following g(x) values guarantees that a single-bit error is caught? $x + 1$, x^3 and 1

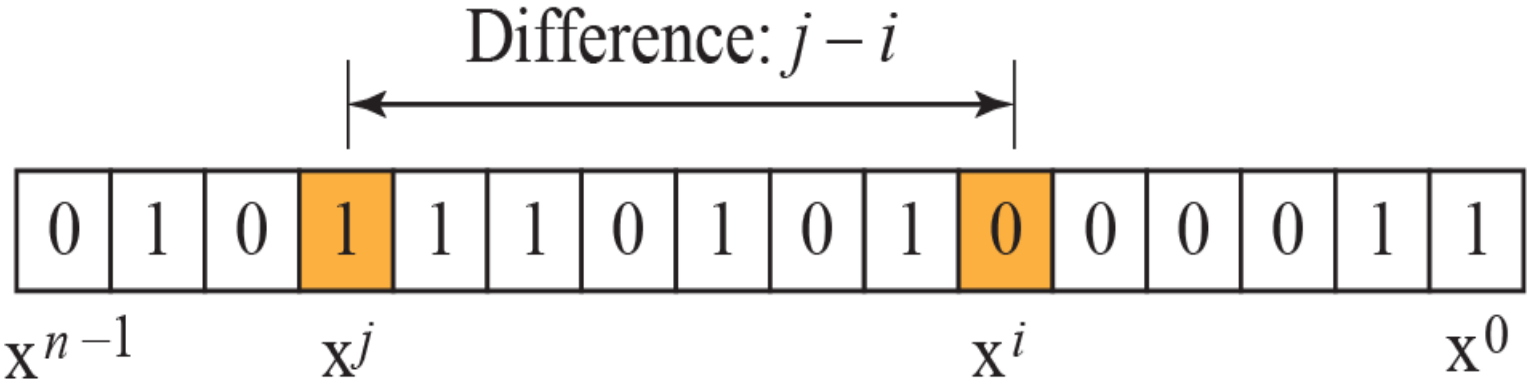
Solution

a. No x^i can be divisible by $x + 1$. In other words, $x^i/(x + 1)$ always has a remainder. So the syndrome is nonzero. Any single-bit error can be caught.

b. If i is equal to or greater than 3, x^i is divisible by $g(x)$. The remainder of x^i/x^3 is zero, and the receiver is fooled into believing that there is no error, although there might be one. Note that in this case, the corrupted bit must be in position 4 or above. All single-bit errors in positions 1 to 3 are caught.

c. All values of i make x^i divisible by $g(x)$. No single-bit error can be caught. In addition, this $g(x)$ is useless because it means the codeword is just the dataword augmented with $n - k$ zeros.

Representation of isolated single-bit errors



Example 10.9

Find the suitability of the following generators in relation to burst errors of different lengths: $x^6 + 1$, $x^{18} + x^7 + x + 1$, and $x^{32} + x^{23} + x^7 + 10$.

Solution

a. This generator can detect all burst errors with a length less than or equal to 6 bits; 3 out of 100 burst errors with length 7 will slip by; 16 out of 1000 burst errors of length 8 or more will slip by.

b. This generator can detect all burst errors with a length less than or equal to 18 bits; 8 out of 1 million burst errors with length 19 will slip by; 4 out of 1 million burst errors of length 20 or more will slip by.

c. This generator can detect all burst errors with a length less than or equal to 32 bits; 5 out of 10 billion burst errors with length 33 will slip by; 3 out of 10 billion burst errors of length 34 or more will slip by.

Example 10.10

Find the status of the following generators related to two isolated, single-bit errors: $x + 1$, $x^4 + 1$, $x^7 + x^6 + 1$, and $x^{15} + x^{14} + 1$

Solution

a. This is a very poor choice for a generator. Any two errors next to each other cannot be detected.

b. This generator cannot detect two errors that are four positions apart. The two errors can be anywhere, but if their distance is 4, they remain undetected.

c. This is a good choice for this purpose.

d. This polynomial cannot divide any error of type $x^t + 1$ if t is less than 32,768. This means that a codeword with two isolated errors that are next to each other or up to 32,768 bits apart can be detected by this generator.

Summary

- A good polynomial generator needs to have the following characteristics:

1. It should have at least two terms.
2. The coefficient of the term x^0 should be 1.
3. It should not divide $x^t + 1$, for t between 2 and $n - 1$.
4. It should have the factor $x + 1$.

Table :Standard polynomials

Name	*Polynomial*	*Used in*
CRC-8	$x^8 + x^2 + x + 1$ **100000111**	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$ **11000110101**	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$ **1000100000100001**	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$ **10000010011000001000111011011 0111**	LANs

Advantages of Cyclic Codes

- Cyclic codes have a very good performance in detecting single-bit errors, double errors, an odd number of errors, and burst errors.

- They can easily be implemented in hardware and software.

- They are especially fast when implemented in hardware which has made cyclic codes a good candidate for many networks.

Other Cyclic Codes

- The cyclic codes we have discussed are very simple.

- The check bits and syndromes can be calculated by simple algebra.

- There are, however, more powerful polynomials that are based on abstract algebra involving Galois fields.

- One of the most interesting of these codes is the Reed-Solomon code used today for both detection and correction.

10-4 CHECKSUM

- Checksum is an error-detecting technique that can be applied to a message of any length.

- In the Internet, the checksum technique is mostly used at the network and transport layer rather than the data-link layer.
- At source,
  - Message is divided into m-bit units.
  - Generator then creates an extra m-bit unit called checksum, which is sent with message.

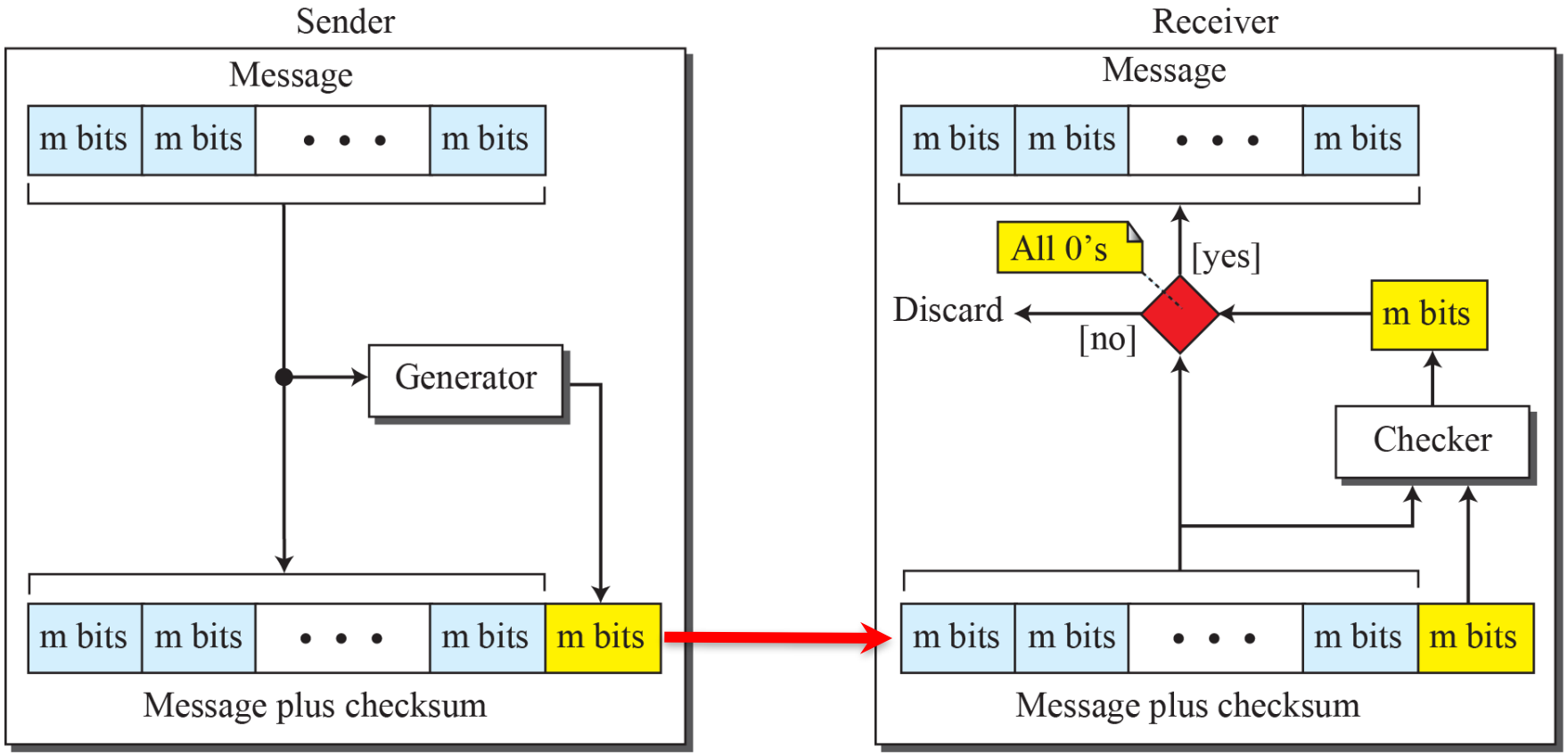
CHECKSUM

- At Destination,

➢Checker creates a new checksum from combination of the message and the sent checksum.

➢If new checksum is all 0's, the message is accepted else discarded.

Checksum


Sender
Message
m bits
m bits
• • •
m bits
Generator
m bits
m bits
• • •
m bits
m bits
Message plus checksum
Receiver
Message
m bits
m bits
• • •
m bits
All 0's
[yes]
Discard
[no]
m bits
Checker
m bits
m bits
• • •
m bits
m bits
Message plus checksum

Concept

- The idea of the traditional checksum is simple.

Show this using a simple example.

Suppose the message is a list of five 4-bit numbers that we want to send to a destination.

In addition to sending these numbers, we send the sum of the numbers.

Eg: if the set of numbers is (7, 11, 12, 0, 6), we send (7, 11, 12, 0, 6, **36**), where **36** is the sum of the original numbers. The receiver adds the five numbers and compares the result with the sum.

If the two are the same, the receiver assumes no error, accepts the five numbers, and discards the sum.

Otherwise, there is an error somewhere and the message not accepted.

Ones compliment Addition

- Each number can be written as a 4 bit word except sum.
- This drawback we can over come using ones compliment arithmetic.
- Represent unsigned numbers between 0 and $2^m - 1$ Using only m bits.
- If the number has more than m bits, the extra leftmost bits need to be added to the m rightmost bits.

Example 10.12

In the previous example, the decimal number 36 in binary is (100100)2. To change it to a 4-bit number we add the extra leftmost bit to the right four bits as shown below.

$$(10)_2 + (0100)_2 = (0110)_2 \rightarrow (6)_{10}$$

Instead of sending 36 as the sum, we can send 6 as the sum (7, 11, 12, 0, 6, 6).

The receiver can add the first five numbers in one's complement arithmetic.

If the result is 6, the numbers are accepted; otherwise, they are rejected.

Example 10.13

Let us use the idea of the checksum in Example 10.12. The sender adds all five numbers in one's complement to get the sum = 6.

The sender then complements the result to get the checksum = **9**, which is 15 − 6.

- Note that 6 = (0110)2 and **9** = (1001)2;they are complements of each other.
- The sender sends the five data numbers and the checksum (7, 11, 12, 0, 6, **9**).
- If there is no corruption in transmission, the receiver receives (7, 11, 12, 0, 6, **9**) and adds them in one's complement to get 15.

Example 10.13

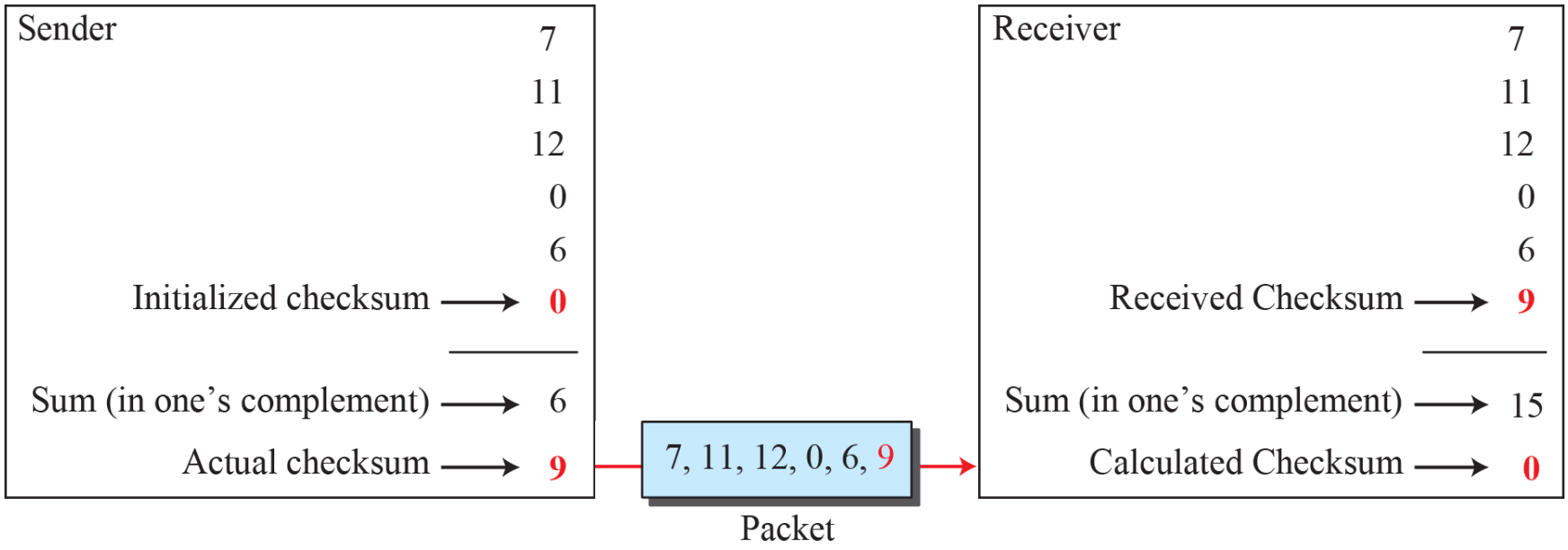


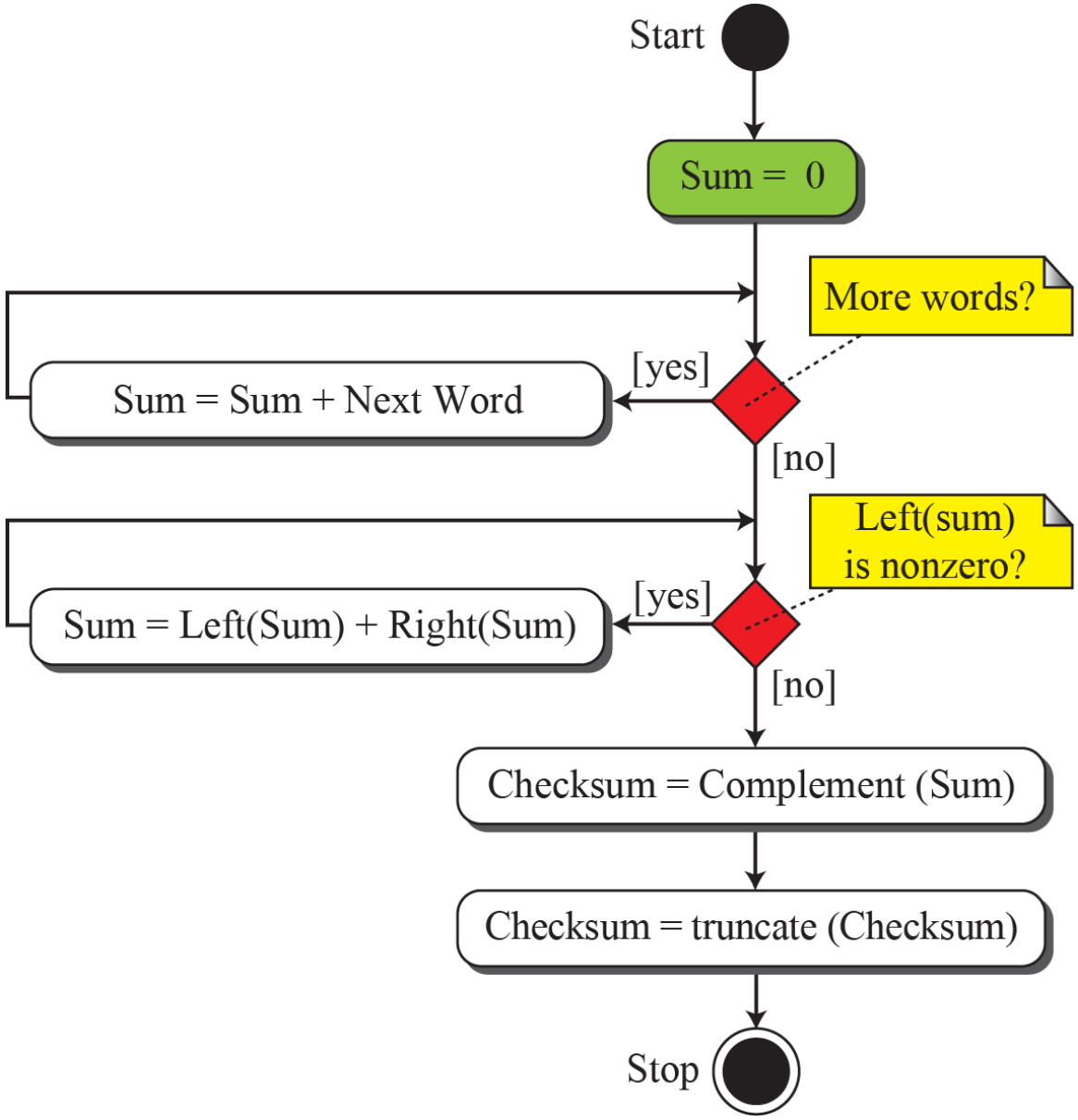
Table : Procedure to calculate the traditional checksum

Sender	*Receiver*
1. The message is divided into 16-bit words.	1. The message and the checksum is received.
2. The value of the checksum word is initially set to zero.	2. The message is divided into 16-bit words.
3. All words including the checksum are added using one's complement addition.	3. All words are added using one's complement addition.
4. The sum is complemented and becomes the checksum.	4. The sum is complemented and becomes the new checksum.
5. The checksum is sent with the data.	5. If the value of the checksum is 0, the message is accepted; otherwise, it is rejected.

Algorithm to calculate a traditional checksum

Notes:

a. Word and Checksum are each 16 bits, but Sum is 32 bits.
b. Left(Sum) can be found by shifting Sum 16 bits to the right.
c. Right(Sum) can be found by ANDing Sum with $(0000FFFF)_{16}$.
d. After Checksum is found, truncate it to 16 bits.



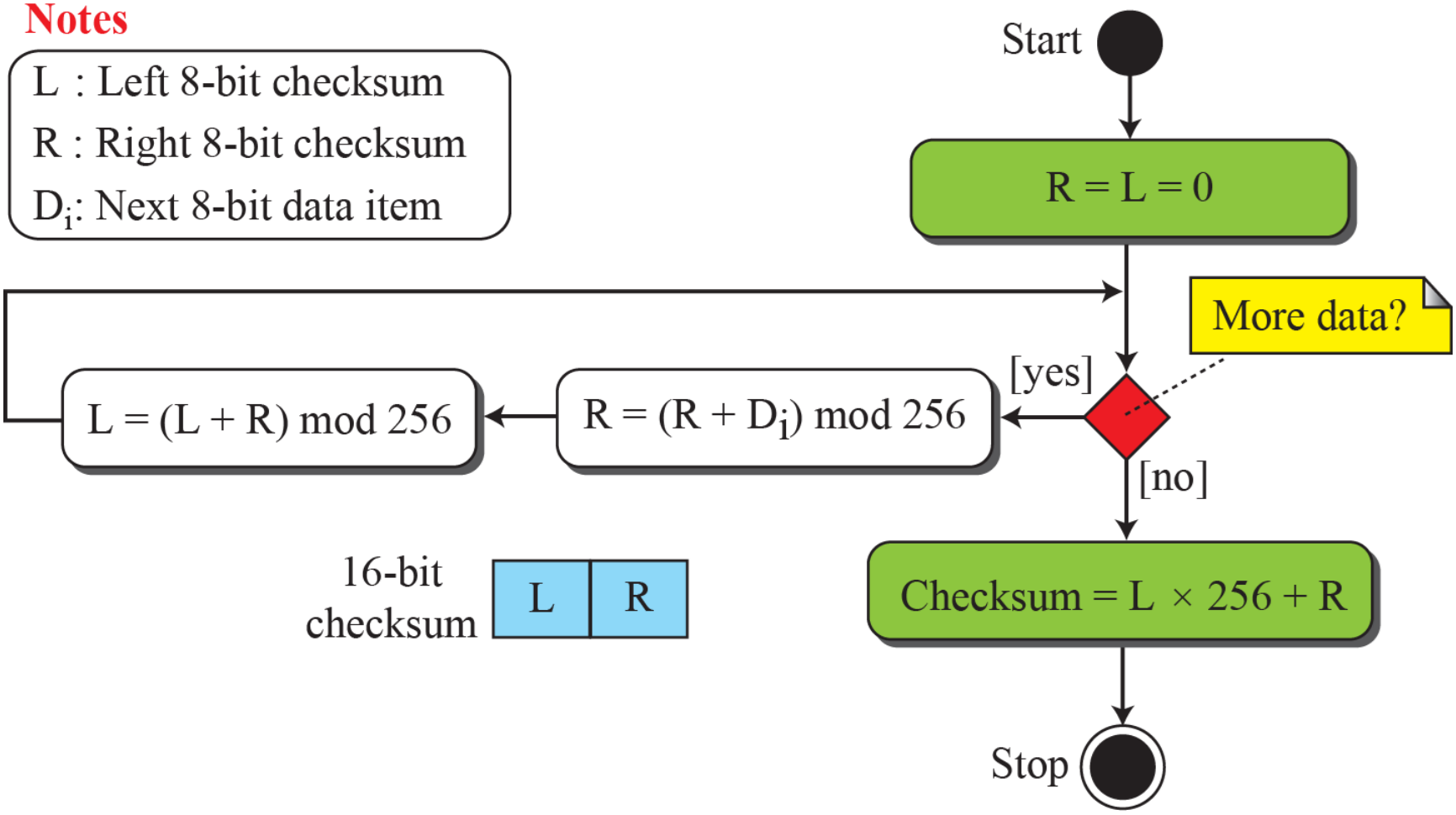
Other Approaches

- One major problem with the traditional checksum calculation.
- If two 16-bit items are transposed in transmission, the checksum cannot catch this error.
- The reason is that the traditional checksum is not weighted: it treats each data item equally.
- In other words, the order of data items is immaterial to the calculation.
- Several approaches have been used to prevent this problem.
- Two of them are: Fletcher and Adler.

Fletcher Checksum

- Was devised to weight each data item according to its position.
- Two algorithms i) 8-bit ii) 16-bit.
- In 8 bit Fletcher, calculation is done on 8-bit data and creates 16 bit checksum.
- Calculation is done using modulo256(2^8)i.e intermediate results are divided by 256 and remainder is kept.
- Algorithm uses two accumulators L & R
- First simply adds data items together,
- Second adds weights to the calculation.
- In 16 bit Fletecher, calculation is done on 16-bit data and creates 32 bit checksum.

Algorithm to calculate an 8-bit Fletcher checksum



Adler Checksum

- Is a 32 bit checksum.
- Similar to Fletcher.
- 3 Major differences

i) Calculation is done on single byte instead of 2 bytes at a time

ii) Modulus is a prime number(65,521)instead of 65,536.

iii) L is initialized to 1 instead of 0

Algorithm to calculate an Adler checksum

